Development of a Mobile Calamansi Juice Yield Prediction System Using *Regressions*

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Chapter 1

This chapter presents the background and context of the study, highlighting the challenges in estimating calamansi juice yield using traditional manual methods and introducing the proposed mobile application that compares three linear regression algorithms Simple Linear Regression, Multiple Linear Regression, and Polynomial Regression to predict the juice yield of calamansi (Citrus microcarpa) based on size and weight. It also outlines the objectives of the study and the scope and limitations of the proposed system.

Introduction

Agriculture plays a significant role in the economic development and food production of many countries, particularly in tropical regions where calamansi-based crops serve as primary sources of income. In the calamansi industry, knowing how much juice a batch of calamansi can produce is a practical concern for juice processors who rely on this information for planning, pricing, and managing their resources. Calamansi (Citrus microcarpa), widely grown across tropical regions, is particularly valued for its juice, which directly affects its market price. Being able to predict juice yield before processing helps juice processors plan their purchasing, production, and distribution more effectively (Xu et al., 2025).

At present, juice processors estimate the expected juice yield of a batch of calamansi from its weight before processing, using an experience-based rule of thumb, while the actual yield is known only after extraction, whether through manual squeezing

or by running the batch through an extraction machine. Because the actual yield varies with the quality of the calamansi, this estimate remains approximate. Manual extraction is also destructive and tedious for large volumes, and doing it for every calamansi wastes both time and produce. There is a need for a data-driven way to estimate yield from measurable physical characteristics before processing begins (Vijayakumar et al., 2023).

Researchers in agriculture have long observed that the physical traits of a calamansi, such as its size and weight, can tell a lot about how much juice it contains. Bigger calamansi tend to produce more juice, and these observable characteristics make it possible to build predictive models that estimate juice content without needing to destroy the calamansi in the process (Yan et al., 2025).

In recent years, machine learning has found its way into agricultural research, offering new tools for improving yield prediction. Among the widely used approaches are variants of linear regression ranging from simple single-predictor models to polynomial models that capture non-linear relationships. These methods are interpretable and practical, making them well-suited for agricultural applications (Hernández et al., 2025).

Studies have shown that prediction systems powered by machine learning can outperform traditional methods in both accuracy and efficiency. By giving juice processors early estimates of how much yield to expect, these systems make planning much easier and more reliable (Yuvasri & Nagarajan, 2024).

In this study, calamansi samples were collected and sorted by size (small, medium, and large), individually weighed, and their juice was extracted and recorded. These data were used to build and compare three regression algorithms: Simple Linear Regression (using weight only), Multiple Linear Regression (using weight and size), and Polynomial Regression of degree 2 (using weight, size, and their interaction and quadratic terms). This comparison allows the study to identify which approach best predicts juice yield based solely on physical characteristics.

The system developed in this study is intended to be a practical tool that juice processors can actually use. Rather than spending time and resources manually squeezing every calamansi just to estimate output, users can enter the total weight of their calamansi batch in kilograms or grams. The system automatically runs all three regression models side by side and presents the comparative results, highlighting Polynomial Regression as the recommended best-performing model.

Purpose and Description

This study was conducted with the goal of designing, developing, and evaluating a mobile application that can predict how much juice a batch of calamansi (Citrus microcarpa) can produce based on the size and weight of the calamansi, by training and comparing three linear regression algorithms.

To do this, calamansi samples were sorted by size into small, medium, and large categories, and each calamansi was individually weighed. Juice was then extracted from the samples and the output measured.

The collected samples were used to train all three regression models: Simple Linear Regression (weight only as predictor), Multiple Linear Regression (weight and size as predictors), and Polynomial Regression of degree 2 (weight, size, and their squared and interaction terms as predictors).

The result is a mobile prediction application built with flutter, deployed as an Android application, where regular users enter the total calamansi batch weight in kilograms or grams. The system converts the value to grams and automatically evaluates the batch across all three regression models (Simple Linear, Multiple Linear, and Polynomial Regression), instantly providing the predicted juice yield in milliliters and liters. The whole approach avoids the need for ripeness classification or image processing, and instead focuses on simple, measurable physical traits that anyone can observe.

The system includes a secure login feature with role-based access control, distinguishing between regular users and an administrator. Regular users such as juice processors can make predictions and view their own prediction history. The administrator has full access to all prediction logs, user management tools, and model performance metrics. All predictions are saved to a database along with a timestamp for future review. User passwords are secured through one-way hashing before being stored, ensuring that account information is protected.

The research followed a quantitative approach, going through the stages of data collection, model development, and system evaluation to make sure the final product is both accurate and usable.

Objectives of the Study

The general objective of this study is to develop a mobile calamansi juice yield prediction application that evaluates and compares three linear regression algorithms Simple Linear Regression, Multiple Linear Regression, and Polynomial Regression (degree 2).

Specifically, this study aims:

1. To collect and categorize calamansi samples into three size groups (small, medium, and large) based on defined physical criteria, and to record the individual weight (in grams) of each sample using a calibrated digital kitchen scale, ensuring a complete dataset with zero missing values during the data gathering phase.

2. To measure the actual juice yield (in milliliters) of each categorized calamansi sample using a standardized manual extraction method and calibrated syringe to establish the ground truth dataset during the experimentation phase.

3. To train and compare three linear regression models—Simple Linear Regression (weight only), Multiple Linear Regression (weight and size), and Polynomial Regression of degree 2 (weight, size, and their squared and interaction terms)—using an 80:20 training-to-test split, targeting an R^2 score of at least 0.80 and the lowest achievable Mean

Absolute Error (MAE) for the best-performing model during the model development phase.

4. To develop a mobile application using Flutter and Dart that accepts the total calamansi batch weight in kilograms or grams, displays the predicted juice yield from the three regression models, and records prediction history, with an administrator module for reviewing user logs and model performance metrics.

5. To evaluate the performance of the mobile application based on prediction response time (target: under 2 seconds per prediction) and the accuracy of its predictions using R^2, MAE, and RMSE.

Scope and Limitations

This study covers the design, development, and evaluation of a mobile calamansi juice yield prediction application built with Flutter. The system was evaluated based on internationally recognized software quality characteristics, specifically focusing on functional suitability, usability, reliability, and performance efficiency.

In terms of functional suitability, the system is scoped to predict the juice yield of calamansi (Citrus microcarpa) using three linear regression variants Simple Linear Regression, Multiple Linear Regression, and Polynomial Regression (degree 2), with size category (small, medium, large) and individual weight (in grams) as the only input features. Regular users enter the total batch weight in kilograms or grams, and the

system converts the value to grams before evaluating the models. To keep the tool rapid and practical in field operations without requiring users to manually count individual samples, the system utilizes the representative unit weight established from the research dataset to evaluate Multiple Linear and Polynomial regressions.

The system performs the functions it was designed for, which is to accurately predict juice yield based on user-provided physical characteristics of calamansi.

In terms of performance efficiency, the system was developed to respond promptly to user inputs and deliver predictions without unnecessary delays. The dataset used to train the models was derived from actual juice extractions conducted during the study period, ensuring that the system operates on real, grounded data.

In terms of usability**,** the system was designed to be simple and accessible for its intended users, juice processors who may not have technical backgrounds. The interface was kept straightforward so that users can input the total weight of a calamansi batch and receive an immediate predicted juice yield without requiring specialized knowledge or equipment. A survey interview with a juice processor was conducted to validate the problem, and User Acceptance Testing (UAT) of the system was likewise conducted; however, the primary focus of this study remained on algorithmic development and technical validation."

In terms of reliability, the system consistently produces predictions based on the trained regression models. However, its predictive accuracy is directly dependent on the

quality and consistency of the collected data, and results may vary across different calamansi varieties or seasonal batches not represented in the dataset.

The study was conducted in Tacloban City, Leyte, Philippines, where calamansi samples were sourced from local market vendors and suppliers. The findings and the developed system are therefore most applicable to calamansi trading and processing activities within this locale and surrounding areas in the region. The dataset covers samples collected within Tacloban City during the study period only, and results may not fully represent calamansi from other regions or growing conditions.

The study does not account for internal factors such as ripeness level or the freshness of the calamansi**,** chemical composition, soil type, or growing conditions, as these fall outside the scope of the measurable physical characteristics targeted by this research. The system was built as an experimental academic prototype and has not been validated for large-scale commercial deployment. Predictions generated by the system should not be used as the sole basis for pricing, purchasing, production, or other financial decisions during research testing or initial deployment. Further refinement and broader data collection would be necessary before wider use.

The model was trained on juice yields obtained by manual extraction, while juice processors typically use extraction machines. Actual yields may therefore differ, and the system's predictions should be treated as estimates. In addition, the system assumes a representative unit weight for the calamansi in a batch, since users are not required to sort calamansi by size.

Additionally, quality characteristics such as security, portability, compatibility, and maintainability were not covered in this study, as they are beyond the scope of its current academic purpose and development stage. However, security measures including password hashing, role-based access control, and secure TLS API communication via Supabase were implemented to protect user accounts and prediction logs, as the system includes user registration and prediction logging functionality.

Chapter 2

This chapter offers a thorough analysis of the literature and associated systems pertinent to the creation of a mobile application that uses regression-based machine learning to forecast the juice production of calamansi (Citrus microcarpa). Regression algorithms (Simple Linear Regression, Multiple Linear Regression, and Polynomial Regression) applied to agricultural datasets, studies on calamansi as a commercially significant agricultural crop in the Philippines, physical calamansi characteristics as predictors of juice yield, and current mobile systems for citrus and crop yield prediction are all covered in this review. The synthesis of these works identifies the gap in readily available, pre-extraction juice yield estimation tools for Filipino calamansi processors, and provides the theoretical and empirical basis for the current study's approach of using batch weight and size category as input features to address.

Review of Related Literature/Systems

2.1 Related Literature

A. Calamansi (Citrus microcarpa) and Its Juice Industry

Abal et al. (2024) conducted a study on the nutritional composition and antioxidant properties of calamansi (Citrus microcarpa) peels subjected to different drying processes, published in Nusantara Bioscience. While their central concern was

the peel byproduct of calamansi juicing, the study confirmed that calamansi grown in the Philippines is primarily cultivated and utilized for its juice, with the juicing process generating the peel as a secondary material. The authors further noted that the Philippines is a significant exporter of calamansi juice to countries such as Japan, South Korea, Hong Kong, the United States, and Canada, underlining the commercial and economic importance of juice yield in calamansi production. This commercial significance directly substantiates the need for a reliable juice yield estimation system such as the one proposed in the current study, particularly for Filipino farmers, vendors, and processors who depend on calamansi juicing for their livelihood.

The bioactive and commercial potential of calamondin (Citrofortunella microcarpa) was reviewed comprehensively by Nguyen et al. (2023), published in PMC/Nutrients. The authors detailed that calamondin, the alternative name for calamansi, is valued precisely for its high juice content, distinctive flavor, and bioactive compound profile including phenolics and flavonoids. Their review established that all parts of the calamondin crop, particularly the juice and pulp, serve applications ranging from ready-to-drink beverages to food ingredients and herbal medicine. The review's emphasis on the calamansi juice as its primary economic derivative validates the focus of the present study on predicting juice yield specifically, rather than total sample weight or other attributes, as juice volume is the most commercially relevant measurable output for stakeholders in the calamansi value chain.

In a study examining the citrus industry in Nueva Vizcaya, Philippines, Idquival et al. (2023), publishing in the International Journal of Research in

Engineering, Science and Management, analyzed production trends, market challenges, and the value chain of citrus farming in the region. Their findings revealed that smallholder citrus farmers in the Philippines continue to face difficulties in estimating the volume of juice they can extract from their harvests a gap that affects pricing decisions, buyer negotiations, and production planning. This contextual finding provides direct empirical support for the development of the calamansi juice yield prediction system in the present study, as it highlights an unmet need for accessible, pre-extraction yield estimation tools among Filipino farmers who do not have access to laboratory-grade equipment or expensive sensing technology.

B. Physical Characteristics of Agricultural Crops as Predictors of Yield and Quality

Sabouri et al. (2025) applied Multivariate Linear Regression (MLR) to non-destructively estimate the weight of produce based on physical dimensions. Their study established that measurable, non-invasive physical dimensions serve as strong predictors of sample weight with R^2 values reaching 0.9369. This finding directly supports the present study's approach of using physical characteristics, specifically weight and size category as reliable predictors of juice yield without destructive extraction. This finding is directly relevant to the present study, which similarly relies on the calamansi weight and size category (small, medium, large) as measurable physical input features, without any image processing or destructive extraction. Both studies share the foundational principle that observable physical traits provide sufficient information for predicting a crop-related quantity of interest.

A study by Basak et al. (2022) applied machine learning models to non-destructively estimate the weight of strawberry samples from physical measurements, achieving high predictive accuracy. The authors demonstrated that when traditional destructive methods are replaced by regression based models trained on measurable features, the result is faster, more scalable, and more practical for large-batch operations. Basak et al. also underscored that multiple regression techniques should be compared when working with agricultural datasets, since the relationship between physical inputs and output quantities may vary in linearity. This methodology directly informs the comparative approach of the present study, which trains and evaluates three regression models Simple Linear Regression, Multiple Linear Regression, and Polynomial Regression to determine which best fits the calamansi juice yield dataset.

Wen et al. (2023) explored the use of non-destructive, affordable in-field measurement to predict the sugariness of a juice-derived quality attribute measured in degrees Brix of strawberries, publishing their findings in Frontiers in Plant Science. The study validated that commodity-level hardware, combined with regression modeling, can provide acceptable prediction accuracy for calamansi juice quality traits without requiring laboratory-grade instrumentation. The authors cautioned that the conventional 'harvest-first, analysis-second' paradigm comes too late for practical preharvest decision-making. This argument mirrors the motivation underlying the current study: by predicting calamansi juice yield from easily measurable physical

characteristics before extraction begins, processors can make more timely and informed decisions about pricing, distribution, and production planning.

Amirat et al. (2021) examined physical and internal crop parameters to assess quality indicators and harvest characteristics using regression analysis. A key finding of their sensitivity analysis was that juice volume accounted for 33% to 50% of the predictive variance in agricultural produce quality assessments, confirming that juice yield is a central, quantifiable informational attribute of citrus and commodities crops rather than merely a secondary byproduct. While their investigation evaluated juice volume alongside other traits, their findings validate the focus of the present study in prioritizing calamansi juice yield as the primary target variable predicted directly from observable physical traits such as calamansi weight and size.

C. Regression Algorithms for Agricultural Prediction

Chicco et al. (2021), writing in PeerJ Computer Science, conducted a comprehensive comparative evaluation of regression evaluation metrics and concluded that R^2 (coefficient of determination) is more informative than MAE, MSE, RMSE, MAPE, and SMAPE as a standard metric for evaluating regression models. The authors argued that R^2 directly indicates how well a model explains the variance in the data, making it the preferred metric when communicating model fit to non-specialist audiences, such as agricultural stakeholders and academic evaluators. The present study adopts R^2 and MAE as its primary evaluation criteria, consistent with this

recommendation, targeting an R^2 of at least 0.80 for the best-performing regression model on the test set.

Piekutowska et al. (2021), publishing in Agronomy (MDPI), compared Multiple Linear Regression (MLR) and artificial neural network (ANN) models for predicting the yield of very early potato cultivars before harvest. Using MAE and MAPE as error metrics, the study found that most models produced MAPE values below 15%, indicating a good level of fit. The ANN outperformed MLR in overall quality metrics, but the authors noted that MLR remained competitive and interpretable, particularly for datasets of moderate size. Their use of an 80:20 training-to-test split mirrors the data partitioning strategy employed in the present study. This study validates the methodological framework of using MLR as a viable and interpretable regression approach for crop yield prediction even when more complex alternatives exist.

Kuradusenge et al. (2023) evaluated regression modeling techniques for agricultural crop yield prediction, using the coefficient of determination (R^2) and Root Mean Squared Error (RMSE) as standard performance evaluation metrics. Their comparative analysis confirmed that Polynomial Regression is an effective predictive model when nonlinear relationships, curvature, or interaction effects exist between predictor features and the target yield. This work provides direct empirical justification for including a 2nd-degree Polynomial Regression model in the present study, testing whether quadratic terms and interaction effects between calamansi sample weight and size categories achieve superior prediction accuracy over standard first-order linear models.

Tanimoto and Yoshida (2024) constructed models for estimating the proportions of citrus size grades using Polynomial Regression, published in *Agronomy*. By applying curvilinear fitting to capture the non-linear relationship between calamansi diameter and size-grade classification, the researchers achieved high estimation accuracy for grading purposes.

This provides direct empirical justification for including Polynomial Regression among the algorithms compared in the present study, since the relationship between calamansi weight and juice yield may likewise exhibit curvature that purely linear models cannot capture. The present study diverges from Tanimoto and Yoshida (2024) in predicting juice volume rather than size-grade distribution, and in directly benchmarking the polynomial model against two simpler regression forms rather than applying it in isolation.

Ansarifar et al. (2021), publishing in Scientific Reports (Nature), introduced an interaction regression model for corn and soybean yield prediction that explicitly incorporated interaction terms between predictor variables. Their model achieved a relative root mean square error of 8% or less across multiple Midwest states, outperforming standard machine learning algorithms. The inclusion of interaction terms between weight and size in the Polynomial Regression model of the current study is conceptually aligned with this approach recognizing that the combined effect of calamansi size and weight on juice yield may not be fully captured by their independent contributions alone.

A review by Lagrazon and Tan (2023) cited in a broader remote sensing yield estimation review established that Multiple Linear Regression (MLR) continues to be one of the most commonly used approaches in RS-based yield estimation, particularly under conditions where relationships are approximately linear but predictor intercorrelations are minimal. Additionally, the review established that when agricultural datasets demonstrate multicollinearity or non-linear growth trajectories, Polynomial Regression offers a more suitable extension by integrating higher-order polynomial terms.

This theoretical framework directly underpins the rationale for the three-algorithm structure of the current study, which tests both linear and polynomial approaches to determine the best fit for calamansi juice yield data.

D. Machine Learning in Agriculture and Yield Prediction Systems

Anbananthen et al. (2021) developed a data-driven decision support system for agricultural yield prediction, establishing a structured mathematical relationship between expected yield output and physical predictor variables. Their findings demonstrated that software-assisted estimation tools provide agricultural stakeholders with fast, reliable, and interpretable yield approximations compared to traditional guesswork. This directly mirrors the design objective of the present study, which develops a mobile calamansi juice yield prediction application allowing users to input total batch weight and immediately receive predicted juice volume in milliliters and liters without destructive manual squeezing.

Yuvasri and Nagarajan (2024) conducted a performance analysis of multiple machine learning models for crop yield prediction in South India using crops such as rice, sorghum, cotton, and sugarcane. The study assessed multiple regression-based models on historical agricultural datasets, confirming that machine learning-based prediction systems consistently outperform traditional experience-based farmer estimation in both accuracy and reliability. The study also noted that farmers traditionally predict crops based on region, soil, weather, and their own experience, which leads to inaccurate estimates particularly under the increasingly unpredictable climate conditions of the present day. This empirical argument directly parallels the motivation of the present study in replacing manual, experience-based juice yield estimation with a data-driven prediction system.

Maheswari et al. (2021), writing in Frontiers in Plant Science, published a comprehensive review of deep learning-based semantic segmentation techniques for crop yield estimation in orchards. The review highlighted that manual yield estimation in agricultural settings is labor-intensive, time-consuming, and imprecise and that intelligent estimation systems offer more benefits to farmers in decisions related to harvesting, marketing, and logistics. While the current study employs regression rather than deep learning, both approaches share the same overarching objective: replacing destructive, labor-intensive yield quantification with a faster and scalable prediction system. The review's finding that estimation systems are beneficial regardless of complexity level reinforces the practical value of even simpler regression-based approaches such as the one implemented in the current study.

Raptis et al. (2025) designed an automated yield estimation tool employing regression analysis to deliver immediate, data-driven yield forecasts for smallholder farmers. Their research emphasized that agricultural software systems must prioritize accessible interfaces and straightforward workflows so that non-technical users can interpret model predictions easily. The present study adopts this user-centered philosophy by embedding regression models into a responsive mobile application built with Flutter, providing local farmers, vendors, and calamansi processors with rapid on-device yield predictions directly on their smartphones without requiring specialized computing hardware.

Rani et al. (2023) developed a smart decision-support system for agricultural planning, with a strong focus on minimizing computational overhead while maximizing practical usability for smallholder farmers who lack access to expensive infrastructure. Their study underscored that systems designed for agricultural end users must prioritize model interpretability, fast response times, and straightforward interfaces alongside technical predictive accuracy. This design principle directly guides the present study,

which employs mathematically transparent regression algorithms implemented within a mobile environment to ensure dependable, instant predictions in low-resource field environments.

A study focused on improving decision support systems with machine learning, published in the Agronomy Journal (Wiley, 2023), identified financial cost and inaccessibility as major barriers to DSS adoption among small farmers. The authors found that free-to-use or subscription-based web platforms significantly reduce

adoption barriers, particularly for smallholder farmers who cannot afford commercial software. They also noted that university-developed academic prototypes serve an important role in bridging the gap between scientific innovation and practical field use. The mobile system developed in the current study designed as an academic prototype accessible without specialized equipment aligns with this recommended approach for maximizing accessibility and uptake among the target agricultural user base.

A tropical crop production modeling review by Castano et al. (2024), published in Frontiers in Agronomy, documented a large surge in publications on machine learning and IoT applications for tropical crop yield modeling beginning in 2021, with rapid acceleration through 2022 and 2023. The review established that 'machine learning,' 'forecasting,' and 'internet of things' are the most recurring keywords in agricultural crop modeling research, and that the data-driven modeling trend is increasingly reaching tropical crop varieties including citrus. The current study is positioned at the intersection of these trends, applying machine learning to calamansi, a tropical citrus crop using a focused and low-technology data collection approach (weight and size measurements rather than satellite or IoT sensors).

Yan et al. (2025), confirmed that physical traits of calamansi can be used to build predictive models that estimate juice content without the need to destroy the produce. The theoretical grounding provided by these two studies directly justifies the present study's decision to use size and weight as sole predictors for juice yield, and confirms that regression-based modeling on physical characteristics is a scientifically valid approach within current agricultural research.

Chicco et al. (2021) and Géron (2022) both established that the 80/20 train-test split is among the most widely adopted data partitioning strategies in machine learning because it provides a large enough training set for the model to generalize while reserving a meaningful test set for unbiased evaluation. The 80:20 split is particularly suitable for moderately sized datasets, such as the one collected in the current study, where maximizing the volume of training data is important for model stability. The present study adopts this split ratio explicitly, with all three regression models trained on 80% of the collected calamansi samples and evaluated on the remaining 20% using R^2 and MAE.

2.2 Related System

A. Citrus and Crop Yield Prediction Systems

Vijayakumar et al. (2023) developed predictive modeling approaches for citrus yield estimation to assist growers in determining expected harvest volume and optimizing crop management. Their study demonstrated that data-driven yield estimation models provide significantly more reliable and scalable predictions than traditional, manual experience-based counting. While their investigation required specialized remote-sensing infrastructure, their findings established the critical commercial value of accessible pre-harvest yield estimation in the citrus sector. The present study addresses this same fundamental need but introduces an accessible, software-driven approach tailored for local calamansi farmers and processors, using

direct, non-destructive physical measurements specifically batch weight and calamansi size deployed on an intuitive mobile interface.

Moussaid et al. (2022), publishing in Informatics (MDPI), developed an intelligent machine learning-based system for predicting citrus harvest yield before the harvest period, using five years of field and satellite data from a Moroccan citrus orchard of 50 parcels. Their system incorporated climate data, irrigation volume, fertilization dose, and parcel size as inputs, and employed Multiple Linear Regression (MLR) and Partial Least Squares (PLS) regression as prediction algorithms. This system is the most thematically proximate to the present study among all related systems, as it applies regression-based modeling specifically to citrus yield prediction. The current study differs in scope focusing on individual-level calamansi juice yield extraction using only weight and size rather than orchard-level tonnage, but draws directly on the regression methodology established in this precedent.

Ebaid et al. (2024), writing in Agronomy (MDPI), employed an artificial neural network (ANN) to predict the yields of five citrus varieties including Valencia orange, Murcott mandarin, and Bearss Seedless lime using weather factors and heat unit accumulation as inputs over 12 growing seasons. Although the prediction target was agricultural tonnage rather than juice volume, the system established that citrus yield prediction from measurable inputs is learnable and practically deployable. The study's methodology, which collected data systematically over multiple seasons and training a model from that data mirrors the data-driven approach adopted in the present study,

which collected juice yield data from physically measured calamansi samples over its study period and used that data to train three regression models.

A machine vision-based citrus grading system described by Li et al. (2025), published in Systems Science & Control Engineering (Taylor & Francis), integrated diameter measurement, coloring rate evaluation, and roundness analysis to perform automated grading of citrus produce in accordance with Chinese national standard GB/T 12947-2008. The system classified citrus into grading categories based on size, color, and shape features. The current study is related to this system in its use of size categories (small, medium, large) as a key classification input, but differs in its purpose the present study uses size as a predictor of juice yield rather than a quality grading output, and does

not require machine vision hardware, making it far more accessible to smallholder farmers.

B. Agricultural Decision Support and Yield Prediction Systems

Anbananthen et al. (2021), publishing in F1000Research, developed a hybrid machine learning decision support system for crop yield prediction accessible via a web page interface. The system compared multiple algorithms including random forest, gradient boosting, and stacked generalization, presenting results instantly to users through a mobile interface. The study's system architecture a web front end that accepts user input, passes it to a trained ML model, and displays the prediction is structurally identical to the system developed in the current study. Both systems also share the

objective of providing fast, reliable, accessible yield estimates to farmers who would otherwise rely on imprecise traditional methods.

Sable et al. (2025) developed an agricultural predictive application that applied regression algorithms to historical agricultural datasets to assist stakeholders in harvest and marketing decisions. Their system evaluated models using standard statistical metrics including Mean Squared Error (MSE) and the coefficient of determination (R^2), presenting predictions directly through an accessible graphical interface. The present study mirrors this comparative methodology by training and evaluating multiple regression models (Simple Linear, Multiple Linear, and Polynomial Regression) using an evaluation matrix (R^2, MAE, RMSE, MSE), allowing users to examine comparative juice yield estimates on a mobile interface.

Raptis et al. (2025) implemented an end-to-end prediction application designed to deliver real-time yield forecasts to agricultural end users. The system translated regression outputs into actionable recommendations through a clean, cost-effective interface tailored for non-specialist users. The present study shares this priority of practical field accessibility by developing an intuitive mobile application that executes natively on Android devices, allowing users to estimate calamansi juice yield instantly from simple physical inputs.

Rani et al. (2023) developed a user-friendly application enabling farmers to enter physical parameters and receive immediate recommendations without requiring technical training. Their system targeted agricultural producers who cannot afford

expensive commercial software, focusing on simplicity and responsive user interface design. While the prediction task differs, the present study incorporates the same design philosophy by translating regression-based prediction models into a clean mobile tool that delivers practical, immediately actionable juice volume predictions for calamansi producers and vendors.

Yuvasri and Nagarajan (2024) built a machine learning-based crop yield prediction system for South Indian crops, comparing multiple models and reporting R^2 values and error metrics. While their system was evaluated primarily as a research prototype rather than as a fully deployed web application, the study included visualizations of model performance that allow users to compare algorithm outputs side by side.

The current study adopts a similar side-by-side comparison display in its Flutter mobile application interface, where all three regression models (SLR, MLR, Polynomial) are compared simultaneously for every user-submitted prediction query.

An intelligent crop recommendation and disease detection system developed by Omara et al. (2023), published in Frontiers in Artificial Intelligence, provided realtime information feedback to smallholder farmers via a mobile interface. The system used machine learning to diagnose crop diseases and offer treatment recommendations based on user-submitted symptom reports, logging all interactions for review. The current study's prediction history logging feature which records all user-submitted predictions with timestamps for later review by both users and administrators is directly inspired by

this approach, recognizing that maintaining a record of predictions enables farmers and processors to track changes in juice yield estimates over time and across batches.

C. Systems with Role-Based Access and Prediction Logging

A study on User Acceptance Testing of an electronic data preprocessing system published in the Jurnal Mahasiswa Ilmu Komputer (2023) evaluated system quality using ISO 9126 dimensions functionality, reliability, usability, and efficiency with a Likert scale instrument, achieving a usability score of 84.25% (Good) and efficiency of 92.00% (Very Good). The study used questionnaires with positive-to-negative gradation options administered to multiple respondents. The current study draw on the ISO 9126 quality dimensions for its evaluation approach, focusing on the model accuracy (R^2 and MAE) and system response time as primary performance indicators, and also conducted a small scale UAT with one respondent (Section 4.5.6)

Adebiyi et al. (2020), publishing in Hindawi Journal of Food Quality, developed a machine learning-aided mobile system for farmland optimization using Random Forest and BigML, deployed through a mobile interface (Appery.io). The system accepted user

inputs such as location, crop type, soil type, and spacing to generate optimization recommendations. The study noted that the mobile/web system needed to accommodate users who have limited technological familiarity. Like this system, the current study's web application is designed to accept non-technical user inputs and immediately return a prediction without requiring the user to understand the underlying mathematical

model. Both systems also include some form of user account management to track interactions.

The crop yield prediction system by Anbananthen et al. (2021), also referenced as a related literature source included a mobile interface with clear user roles, distinguishing between the farmer/end-user who submits inputs and the researcher/administrator who reviews model outputs and performance metrics. This dual-role design is reflected in the current study's system, which implements role-based access control (RBAC) distinguishing between regular users (farmers, vendors, processors) who make predictions and view their own prediction history, and an administrator who has full access to all prediction logs, user management tools, and model performance metrics.

A crop damage prediction framework published in Scientific Reports (Waltz et al., 2025) proposed the future development of a user-friendly decision support system to empower farmers with accessible tools providing actionable insights, recommendations, and visualizations.

Their system framework emphasized the importance of role-separated access distinguishing operational users from system administrators and highlighted prediction logging as an essential feature for monitoring model performance over time. The current study's system design directly implements these recommendations, incorporating both role-based access control and prediction history logging with

timestamps, supporting the administrators' ability to monitor and review the system's outputs in real-world use.

D. Software Architectures and Frameworks for Agricultural Prediction Applications

Akkem et al. (2024), cited in Scientific Reports, developed a Streamlit-based crop recommendation system with advanced Explainable Artificial Intelligence (XAI) for smart farming, published in Neural Computing and Applications. Their system used Streamlit to create an accessible web front end that allowed farmers to receive crop recommendations with explanations for why a particular crop was suggested. The study confirmed that Streamlit is among the most suitable frameworks for deploying machine learning models in agricultural contexts because it requires minimal front-end coding expertise and can be maintained by researchers without full-stack web development skills. The present study adopts these design principles of minimal complexity and responsive UI components, implementing them within a Flutter mobile environment to maximize field portability for local growers.

Waltz et al. (2025), publishing in Frontiers in Artificial Intelligence, developed a cyberinfrastructure for machine learning applications in agriculture at The Ohio State University, processing one terabyte of multimodal data (UAV imagery, soil sensors, weather sensors) collected during the 2023 growing season. The paper discussed the challenges of building scalable ML agricultural systems that remain accessible to different user types. While this system is far more complex than the one in the current study, it reinforces the principle that mobile interfaces serving multiple stakeholder

types (researchers, farmers, extension agents) require role-differentiated access structures a principle implemented in the current study through its RBAC architecture distinguishing regular users from administrators.

A system developed for precision agriculture by Rani et al. (2023) in Scientific Reports included a mobile interface designed specifically to minimize the technical knowledge required for agricultural decision-making. The system's quantitative results, achieved through 10-fold cross-validation and metrics including accuracy and F1score,

demonstrated that even relatively simple models can achieve high performance when carefully designed and evaluated. The current study shares this philosophy prioritizing simplicity, interpretability, and practical deployment while using regression-specific metrics (R^2 and MAE) as its evaluation benchmarks.

Sable et al. (2025) highlighted the practical necessity of historical prediction logging in agricultural applications, enabling users to review past estimates and analyze trends across multiple batches over time. Their system stored query records linked to user accounts for longitudinal review. The present study integrates an identical prediction history feature, automatically saving every batch weight entry, algorithm utilized, timestamp, and predicted juice volume to a cloud database (Supabase), providing farmers, vendors, and administrators with a complete, traceable audit trail.

E. Security, Usability, and Evaluation of Agricultural Systems

Provos and Mazieres (1999), widely cited in subsequent cybersecurity literature, introduced the bcrypt password hashing algorithm as a more secure alternative to

traditional cryptographic hash functions. By 2023, a study published through USENIX on the 25-year retrospective of bcrypt confirmed that the algorithm remains one of the most widely deployed password hashing mechanisms in web applications because its work factor can be adjusted over time to keep pace with increasing computational power. The present study uses one-way password hashing before storing user credentials in the database, consistent with this established best practice for protecting user account security in mobile systems.

A study on secure hashing in web applications published in IEEE proceedings (2023) found that 48.94% of surveyed web application frameworks use BCRYPT hashing, a positive trend toward more secure password storage confirming its widespread adoption across application domains. The study compared BCRYPT, SHA256, and MD5, finding that BCRYPT consistently outperforms older hashing algorithms in resistance to brute-force and rainbow table attacks. The secure password storage implemented in the current study's system, using one-way hashing of user passwords before database storage, is grounded in these industry best practices documented in the cybersecurity literature.

A UAT survey report conducted by TestMonitor (2023) across software development professionals found that UAT remains the final, most critical quality validation stage before software release, ensuring that a system works as expected for real-world end users.

The survey also confirmed that Likert scale questionnaires remain the most common instrument for measuring satisfaction and usability in UAT processes. While

the primary focus of the current study is technical validation through model accuracy metrics (R^2 and MAE) and system response time, a small-scale UAT with one respondent was also conducted (Section 4.5.6), and this literature supports wider user-centered evaluation as a recommended direction for future work.

A study on cyberinfrastructure for agricultural machine learning by Waltz et al. (2025) emphasized that agricultural machine learning systems must be validated not only for technical performance metrics but also for their usability with intended enduser populations including farmers and extension workers. The authors argued that a system achieving high model accuracy but failing in user usability metrics provides less real-world value than a well-designed, accessible system with moderate prediction performance. The current study addresses the technical validation component through model accuracy evaluation (R^2 and MAE) and system performance testing (response time), while acknowledging that larger-scale usability testing with more end-users is a recommended direction for future research and development.

A field-based crop disease recommender system by Omara et al. (2023), published in Frontiers in Artificial Intelligence, developed a web-accessible system for smallholder farmers in Uganda that combined machine learning with a natural language interface. The system was validated through user testing with farmers who provided feedback through structured questionnaires.

The study found that even users with low technology literacy could successfully use the system after minimal onboarding, demonstrating that well-designed agricultural web applications can achieve high usability scores with

non-technical users. The current study's calamansi juice yield prediction system is similarly designed for non-technical users farmers, vendors, and processors with a simple input interface that requires no prior knowledge of machine learning or statistical methods.

Chapter 3

This chapter explains the technical concepts, tools, and technologies used in the development of the system. It includes discussions on the three linear regression algorithms used, system architecture, software and hardware components, and other technical elements necessary for predicting calamansi juice yield.

Technical Background

Current System

Calamansi juice processing in the Philippines have long relied on manual, hands-on methods to assess and manage calamansi yield. In the current practice, juice processors have no reliable way to estimate how much juice a batch of calamansi will produce before it is processed. The entire process begins with a visual inspection of the calamansi, where handlers sort the calamansi by size.This sorting is informal and inconsistent, with no standardized criteria applied across different handlers or locations.

Once sorting is done, juice yield is determined only after extraction, either manually or by machine. In manual extraction, each calamansi is cut open and squeezed by hand or with a basic handheld juicer, and the juice is measured using ordinary kitchen tools such as cups or bowls, which are not always calibrated accurately. Some processors use an extraction machine, which is faster but still gives the result only after the batch has been processed. In both cases, the yield is known too late to guide purchasing, pricing, and production planning, and calamansi used for

manual yield estimation cannot be sold or used for other purposes, which adds up to significant waste especially in larger batches.

Pricing and processing decisions are also largely based on intuition in the current setup. Without a reliable yield estimate, juice processors struggle to plan their production volumes accurately. There is no system in place that records or tracks juice yield data over time, which means processors cannot use past information to improve their decisions. All records, if kept at all, are written manually on paper or simply remembered, making it difficult to identify patterns or refine practices.

These limitations highlight a clear gap in the current system. The lack of a structured, data-driven approach to estimating calamansi juice yield leads to inefficiencies, waste, and missed opportunities for better planning. There is a need for a smarter and more accessible tool that can give processors a reliable prediction of juice yield before extraction begins, without requiring them to destroy any of their produce in the process.

Proposed System

The proposed system is a Calamansi Juice Yield Prediction Mobile Application developed using the Flutter framework and Dart. It trains and compares three regression algorithms to predict the volume of juice measured in milliliters that can be extracted from a given batch of calamansi. For regular-user predictions, the system uses the total batch weight as the only required input. The weight may be entered in kilograms or grams and is converted to grams before running the prediction engine, which outputs

side-by-side estimates from Simple Linear, Multiple Linear, and Polynomial Regression models while highlighting Polynomial Regression as the optimal model. The system was trained on a dataset of real calamansi samples that were physically weighed using a digital kitchen scale and juiced during the data collection phase of this study, with juice volume measured using a standard syringe.

When a user submits their batch weight, the system converts the value to grams and evaluates the batch against all three regression models using the representative empirical unit weight. The system then produces the predicted juice yield in milliliters and liters for each model. All predictions are saved to a database along with a timestamp and can be reviewed later through the prediction history feature.

The system includes role-based access control, distinguishing between regular users and administrators. Regular users can make predictions and view their own history, while administrators have access to all prediction logs, user management tools, and model performance results. User passwords are secured using SHA-256 hashing before being stored in the database.

Table1:

Software

Name of Software	Features Used in Project Development	Purpose
Python 3.x	General-purpose programming language with extensive libraries for data science and machine learning	Serves as the core backend language for data processing, model training, and system logic
Flutter SDK (Dart)	Open-source UI software development kit for crafting natively compiled mobile applications from a single codebase	Used to build the responsive mobile user interface where users input batch weights, view side-by-side regression results, and manage prediction history
scikit-learn	Python machine learning library providing tools for model training, evaluation, and prediction	Used to train and deploy all three regression models for juice yield prediction
Supabase (PostgreSQL)	Cloud-hosted relational database with built-in authentication, role-based access control, and real-time synchronization	Securely manages user authentication credentials, role assignments, and centralized prediction history logs
pandas	Python library for data manipulation and analysis using DataFrame structures	Used to load, clean, and process the calamansi dataset during model training
NumPy	Fundamental Python library for numerical computation and array operations	Supports mathematical calculations in feature engineering and prediction processing
joblib	Python library for lightweight pipelining and model serialization	Saves and loads the trained regression models as .pkl files for real-time predictions
hashlib (SHA-256)	Python built-in library for cryptographic hashing functions	Secures user passwords through one-way SHA-256 hashing before storing in the database

Visual Studio Code	Lightweight source code editor with built-in Git, debugging, and extension support	Primary development environment used for writing, testing, and debugging the system code

Dart and the Flutter framework served as the core development environment for the user-facing mobile application, enabling on-device mathematical regression calculations and fluid UI interactions on Android devices.Supabase provides the backend authentication and relational cloud database services.

For the machine learning component, scikit-learn provided implementations for all three regression algorithms. Simple Linear Regression and Multiple Linear Regressions were implemented using scikit-learn's LinearRegression class. Polynomial Regression (degree 2) was implemented using a Pipeline that combines PolynomialFeatures (degree=2, include_bias=False) with LinearRegression, allowing the model to automatically generate squared and interaction terms from the weight and size inputs. pandas and NumPy were used together to load, clean, and process the dataset, as well as to handle numerical computations involved in feature engineering.

Supabase (PostgreSQL) was chosen as the database solution to provide scalable, cloud-synchronized data storage for user accounts and prediction logs across mobile devices. On-device local storage via SharedPreferences was also integrated to ensure rapid offline history retrieval. Password security is handled through Python's built-in hashlib library using SHA-256 hashing, ensuring that user passwords are never stored

in plain text. Visual Studio Code was the development environment used throughout the project for writing, testing, and debugging the code.

Table 2:

Hardware

Name of Hardware	Features Used in Project Development	Purpose
Smartphone / Tablet	Android mobile operating system, touch screen display	Allows end users such as juice processor to access the mobile prediction application in the field.
Personal Computer / Laptop	Minimum 4GB RAM, dual-core processor, internet connectivity, standard keyboard and display	Used for developing, running, and testing the system and training the prediction model
Digital Kitchen Scale	Digital display, portable, used for measuring weight in grams	Used to measure the individual weight of each calamansi sample in grams during data collection
Manual Hand-Press Juicer	Heavy-duty lever-action hand-press citrus squeezer with perforated extraction cup	Used to extract juice from each calamansi sample with uniform mechanical pressing.

Standard Syringe	Marked in milliliters, transparent barrel with visible measurement markings	Used to measure the volume of juice extracted from each calamansi sample in milliliters

A personal computer or laptop was the primary hardware used throughout the development of the system. It was used to write and run the code, train the machine learning model, and test the system. A dual-core processor and a minimum of 4GB RAM were sufficient to handle the computations involved, given the relatively small size of the dataset.

During the data collection phase, a digital kitchen scale was used to measure the weight of each calamansi sample in grams. Accurate weight measurement was critical since weight is one of the two main input variables. Each calamansi was then placed into a manual hand-press juicer and the juice was extracted under uniform pressure. A standard syringe was used to measure the exact volume of juice obtained from each sample in milliliters. This simple extraction method ensured that the data collected was consistent and reliable enough to train an accurate model.

On the user side, the application is deployed directly to mobile handheld devices (smartphones and tablets running Android). This ensures that processors in local trading posts and field harvest areas can execute predictions directly on their personal mobile devices.

Peopleware

The proposed system involves the following users, all of whom must access the system through log-in credentials:

1. System Administrator - has full access to the system and is responsible for managing user accounts and maintaining the database records. The administrator also oversees the overall performance of the system and monitors all prediction logs generated by regular users. In addition, the administrator manages user roles, can remove accounts, and reviews the model performance dashboard showing evaluation metrics such as R^2 score and Mean Absolute Error to ensure the system is functioning accurately and

reliably.

2. User (Juice Processor) refers to the primary end users of the system. After logging in, regular users enter the total weight of their calamansi batch in kilograms or grams. The system automatically computes predictions for all three regression models, presenting the outputs with the best-performing Polynomial model highlighted. Regular users can also view and track their prediction history. By using the system, the juice processor can estimate how much juice a batch of calamansi will yield before processing. This helps in deciding how much calamansi to buy, setting fairer prices, and planning production. It also saves the time and calamansi that would otherwise be used for test squeezing, and the prediction history allows the processor to review past estimates to improve future decisions.

3. System Developer / Maintenance Personnel - are responsible for the initial development, deployment, and ongoing maintenance of the system. They handle the installation and configuration of the system, perform updates and bug fixes, and optimize the regression models and system components as needed. They also ensure that the database, prediction logic, and user interface continue to function correctly over time and address any technical issues that may arise during system operation.

Security Measures

The system implements fundamental security measures to protect user accounts and data:

1. User Authentication & Password Security – All users must create an account and authenticate using credentials.Passwords are hashed using SHA-256 before storage in the database and are never stored in plain text.
2. Role-Based Access Control (RBAC) – The system differentiates between regular user (processors) and administrators. Regular users can only access their own prediction history, while administrative privileges are restricted to authorized personnel for monitoring system logs and model performance metrics.
3. Input Validation – All user inputs (such as weight in kilograms or grams) are validated and sanitized in real time. The system prevents negative, zero, or non-numeric entries, ensuring erroneous data does not affect regression calculations.

Chapter 4

This chapter presents the research methodology used in developing the Mobile Calamansi Juice Yield Prediction System Using Regressions, including data collection, dataset preparation, system design, and model training. It also discusses the results obtained from the system testing and evaluation, as well as the analysis and interpretation of the prediction accuracy and system performance.

METHODOLOGY, RESULTS, AND DISCUSSION

4.1 REQUIREMENTS ANALYSIS

Requirement's analysis is an important phase in system development that identifies the necessary functional and non-functional requirements needed for the system to operate effectively. It defines what the system must do and the conditions under which it must operate. For the proposed Development of a Mobile Calamansi Juice Yield Prediction System Using Regressions, the requirements focus on ensuring that the system performs efficiently, safely, and securely.

The requirements were identified based on the system's objectives, expected users, and operational environment. These requirements serve as guidelines for system developers to ensure that the developed system meets the needs of users while maintaining reliability and security.

Table 3: Performance Requirements of the Proposed System

Requirements	Description	Rationale
Response Time	The system must generate juice yield predictions within 2 seconds of user input submission.	Fast response time ensures usability for juice processor who need immediate feedback during sorting and pricing activities.
Prediction Accuracy	The system should generate accurate juice yield predictions using the best-performing regression model based on weight and size.	Accurate predictions provide reliable results that users can trust for production planning and pricing decisions.
Input Processing	The system must correctly accept and process user inputs for total batch weight in kilograms or grams and display comparative yield results across all regression models.	Proper input handling ensures that the regression model receives correctly formatted data, preventing erroneous predictions.
Throughput	The system should handle up to 100 prediction queries per session without performance degradation.	Vendors and processors often evaluate multiple batches in a single session, requiring reliable and consistent throughput.

System Responsiveness	The system interface should respond to user inputs within 1–2 seconds.	Quick response time enhances usability and provides smooth interaction for users.

The performance requirements highlight the need for a highly responsive prediction system capable of generating juice yield estimates based on user-provided inputs. Prediction accuracy is critical because each misprediction affects the reliability of juice yield estimation, which is essential for juice processors making production and pricing decisions. Throughput ensures that the system can handle multiple batch queries within a single session without slowing down. System responsiveness guarantees that predictions appear almost instantaneously, ensuring seamless user interaction and maintaining confidence in the system's outputs.

Table 4: Safety Requirements of the Proposed System

Requirement	Description	Safeguard
Input Validation	The system should verify that all user inputs are within acceptable ranges before processing them.	Prevents invalid or out-ofrange values from being passed to the regression model, avoiding erroneous predictions.

Error Handling	The system should identify and manage errors during input processing or prediction generation.	Helps maintain system stability and notifies users when inputs are missing, invalid, or outside acceptable ranges.
Data Integrity	The trained regression model parameters and dataset must be protected from accidental modification.	Protects the integrity of all three prediction model, ensuring that predictions remain accurate and consistent.
Controlled Processing	The system should limit input fields to their defined data types and value ranges.	Prevents excessive or incorrect data entry that could cause processing errors or incorrect model outputs.

The safety requirements are crucial for smooth system operation and preventing disruptions. Input validation ensures the system accepts only properly formatted data, preventing errors during the prediction process. Error handling mechanisms allow the system to continue functioning even when unexpected problems occur, such as missing fields or out-of-range values, by detecting issues and notifying the user. Data integrity measures protect the regression model and dataset from accidental modification, maintaining the accuracy and consistency of predictions. Controlled processing limits

inputs to their defined data types and value ranges, preventing incorrect data entry from causing processing errors or inaccurate model outputs.

Table 5: Security Requirements of the Proposed System

The security requirements are paramount in modern software systems, especially those handling data and machine learning models. Password security through SHA-256 hashing ensures that even if the database is compromised, user credentials remain protected. Secure storage of all three regression model

Requirement	Description	Implementation
Data Privacy	The system shall store only prediction inputs and outputs tied to user accounts. No sensitive personal information beyond username is collected.	All inputs and outputs are stored in the database linked to the user's account. No additional personal data is collected.
Secure Data Storage	The trained regression model and dataset must be stored securely and not exposed to end users through the interface.	Store model files in protected server-side directories inaccessible through the mobile application interface
Access Control	Only administrators should be able to access all prediction logs, manage users, and view model performance metrics.	Role-based access control separates user and admin dashboards. Users can only view their own prediction history.
Secure Deployment	The application shall be deployed using HTTPS to encrypt data in transit between the mobile application and the server.	Hosted on Supabase with TLS/HTTPS encrypted connections to protect user data and prediction logs during transmission between the mobile app and cloud database.

files in protected server side directories maintains their integrity and prevents unauthorized access. Access control mechanisms ensure that only administrators can view all prediction logs, manage users, and inspect model performance dashboards.

4.2 Design of Software, Systems, Product, and/or Processes

4.2.1 System Development Life Cycle (SDLC)

The development of the proposed system followed the Iterative Software Development Life Cycle (SDLC) Model. The iterative model is a development approach in which the system is built and improved through repeated cycles or iterations. Each iteration involves planning, designing, implementing, and testing a portion of the system. After each cycle, the system is evaluated and refined before moving to the next iteration. Unlike traditional linear development models, the iterative model allows developers to gradually enhance the system through continuous improvements. This approach is particularly useful for projects involving machine learning and statistical modeling, where system performance can be improved through repeated testing and adjustments. In this study, the iterative model allowed the researchers to progressively improve the accuracy of all three regression models. Each iteration involved refining the feature engineering process, comparing model performance metrics, and testing the prediction results against the

defined accuracy thresholds. The comparison of three algorithms (Simple LR, Multiple LR, and Polynomial Regression) was itself an iterative evaluation to identify the best-performing approach.

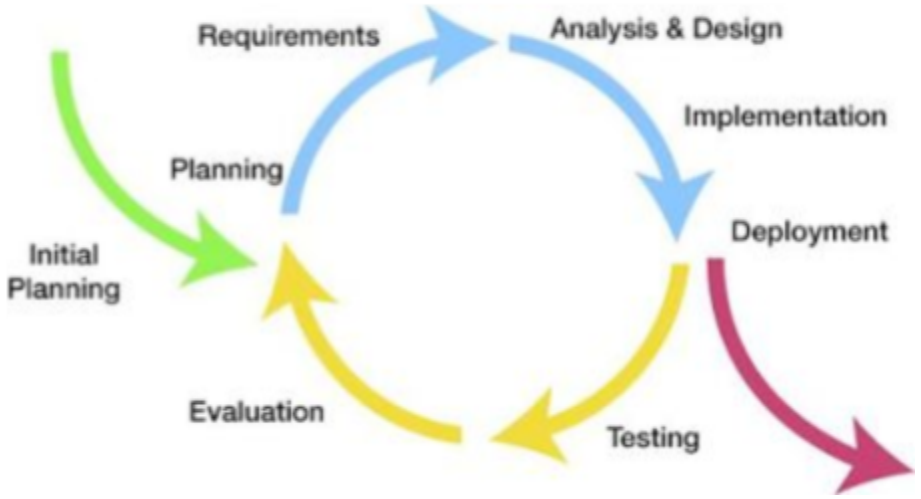


Figure 1. *Iterative Model of the Study*

1. Initial Planning – The initial planning phase marks the beginning of the system development process. In this stage, the researchers identified the main problem that the study intends to address. The traditional method of measuring calamansi juice yield involves physically extracting juice from each calamansi using a hand-press juicer and standard syringe, which is a destructive and time-consuming process. This limitation motivated the researchers to explore a data-driven approach that could estimate juice yield from observable physical characteristics, specifically calamansi weight and

size without damaging the calamansi. During this phase, the objectives of the research were clearly defined, and the feasibility of comparing three regression algorithms on the collected calamansi dataset was established.

2. **Planning** – The planning phase involves organizing the activities required to complete the project successfully. During this stage, the researchers developed a structured timeline outlining the tasks needed to implement the proposed system. Key activities identified in this phase included collecting and measuring calamansi samples, designing the system architecture, implementing data preprocessing steps, training all three regression models, and conducting system testing. The researchers also identified the hardware and software tools required for system development, including the digital kitchen scale for weight measurement and the standard syringe used for juice volume measurement during data collection.

3. **Requirements** – The requirements phase focuses on identifying the functional and non-functional requirements. Functional requirements include allowing regular users to enter the total weight in kilograms or grams, converting the value to grams, automatically applying all three regression models (Simple Linear, Multiple Linear, and Polynomial), displaying the predicted juice yield, and saving the result to the user's

prediction history. Administrators can manage users, view all prediction records, and review all three trained models.

4. **Analysis and Design** – The analysis and design phase focuses on developing the system architecture and designing the workflow of the system. Several system diagrams were created to illustrate how the system components interact. The researchers also designed the data preprocessing pipeline, which includes input validation, conversion of the batch weight using the representative unit weight, and prediction modeling for all three algorithms.

5. **Implementation** – The implementation phase involves translating the system design into an actual working system. The system was developed using Python. scikit-learn was used to implement all three regression algorithms. Simple and Multiple Linear Regression were implemented using LinearRegression. Polynomial Regression (degree2) was implemented using a Pipeline combining Polynomial Features andLinear Regression. All three models were calibrated and integrated into the application service layer. The Flutter mobile interface computes and displays a side-by-side comparison of all three regression predictions.

6. **Testing** – Testing is conducted to verify that the developed system operates according to the defined requirements. White-box testing was conducted to verify individual functions including input

parsing and prediction pipeline operation for all three models. Black-box testing assessed functional suitability, usability, performance efficiency, and reliability based on internationally recognized software quality characteristics. The performance of all three regression models was evaluated using Mean Absolute Error (MAE) and the coefficient of determination (R^2).

7. **Deployment** – Deployment refers to the process of making the system available for use by the intended users. In this study, the developed system was packaged as an Android application package (APK) using Flutter, allowing users to install and run the prediction system natively on handheld mobile devices.

8. **Evaluation** – The evaluation phase assesses the overall performance of the system after testing and deployment. The researchers reviewed prediction results across all three algorithms and compared their metrics. If system performance was not satisfactory, adjustments were made including retraining models with different hyperparameters or refined data splits.

Table 6: Test Plan

Test Case	Description	Expected Result	Assessment
Input Validation Test	Submit missing, zero, negative, and out-of-range weights using both kilogram and gram units.	The system displays descriptive error messages and does not generate a prediction.	Verify that all invalid inputs are caught before reaching any regression model.
Prediction Accuracy Test	Submit known calamansi sample data and compare system output with actual measured juice yield.	Predicted values fall within acceptable error margins ($R^2 \geq 0.80$ target for the best model; achieved $R^2 = 0.7102$, see Section 4.5.2).	Evaluate prediction accuracy using R^2 and MAE for all three models.
Model Integration Test	Submit valid weights in kilograms and grams and verify that the values are converted correctly and processed by each of the three regression models (Simple Linear, Multiple Linear, and Polynomial).	End-to-end pipeline processes inputs and returns a prediction without errors.	Verify that Simple LR, Multiple LR, and Polynomial Regression each produce independently correct outputs.
Interface Test	Test the Flutter mobile application interface for usability and responsiveness.	Interface responds to user inputs within 1–2 seconds and displays results clearly.	Verify that the interface responds to user inputs within 1–2 seconds and displays results clearly.

Administrator Model Analysis Test	Verify that administrators can access the performance results of Simple Linear Regression, Multiple Linear Regression, and Polynomial Regression.	Administrator dashboard displays R^2 and MAE metrics for all three trained models with visual comparisons.	Confirm administrator has access to all model performance dashboards and can view training and test metrics for each algorithm.

The test plan is designed to validate every stage of the prediction workflow. Input validation testing ensures the system correctly handles missing or out-of-range data before it reaches the regression model. Prediction accuracy testing confirms that the model produces reliable juice yield estimates by comparing system outputs against actual measured values from the dataset. Integration testing verifies the complete data pipeline from user input through preprocessing and model inference to output display. Interface testing validates usability, ensuring that results are displayed clearly and without delay. This comprehensive testing approach ensures the system operates efficiently and meets user expectations.

Table 7: Security Plan

Security Measure	Description	Purpose
User Authentication	All users must log in with credentials. Administrators have a separate login path.	Use secure login credentials to enforce role-based access control.

Password Hashing	All passwords are hashed using SHA-256 before storage. Plain text passwords are never stored.	Ensures user passwords remain secure even if the database is accessed.
Model Protection	Regression model .pkl files are stored server-side and are not accessible via the UI.	Prevents unauthorized access to or manipulation of the prediction model.
Input Sanitization	All user inputs are validated and sanitized before being passed to the model.	Prevents injection attacks and protects against erroneous model inputs.

The security plan reinforces the protection of both user data and model assets. Since the system processes agricultural data, data privacy measures ensure that no personal information is stored or transmitted beyond what is necessary for the system to function. Model protection mechanisms maintain the integrity of all three regression models by preventing direct access to their parameters through the interface. Input sanitization protects the system against malformed data entries that could produce incorrect predictions or cause processing errors.

Table 8: Maintenance Plan

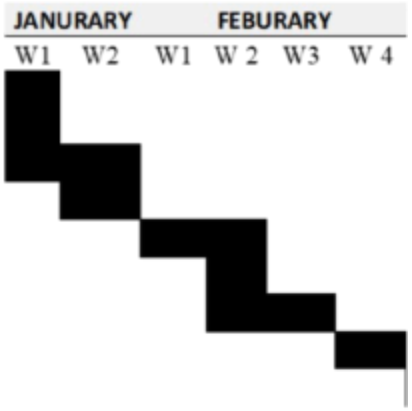
Maintenance Activity	Description	Frequency
Dataset Update	Add new calamansi samples from additional harvest seasons to improve model training.	Periodically, per harvest season.
Model Retraining	Retrain all three regression models with updated data to maintain prediction accuracy and verify which algorithm remains best-performing.	When the dataset is expanded or prediction error exceeds acceptable threshold.
System Fixing	Improve system performance, fix bugs, and update Python library dependencies.	As needed, upon discovery of issues or quarterly dependency review.

The maintenance plan ensures the long-term reliability of the prediction system. Periodic dataset updates allow all three regression models to adapt to calamansi variations across different harvest seasons, improving prediction accuracy over time. Model retraining ensures that predictions remain accurate as the dataset expands, accounting for differences in calamansi characteristics across growing conditions. System fixing addresses software issues that may arise during operation, while dependency updates keep the application

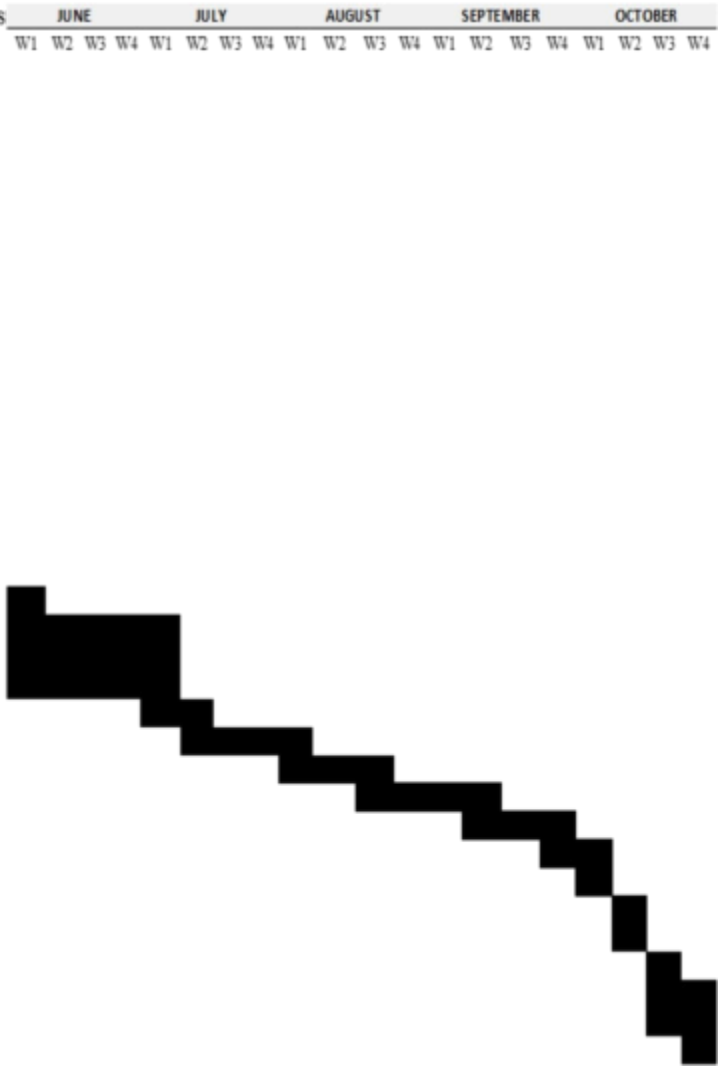
compatible with the latest versions of its core libraries. This proactive maintenance strategy ensures the system remains robust and accurate in real-world, continuous-use scenarios.

4.2.2 GANTT CHART

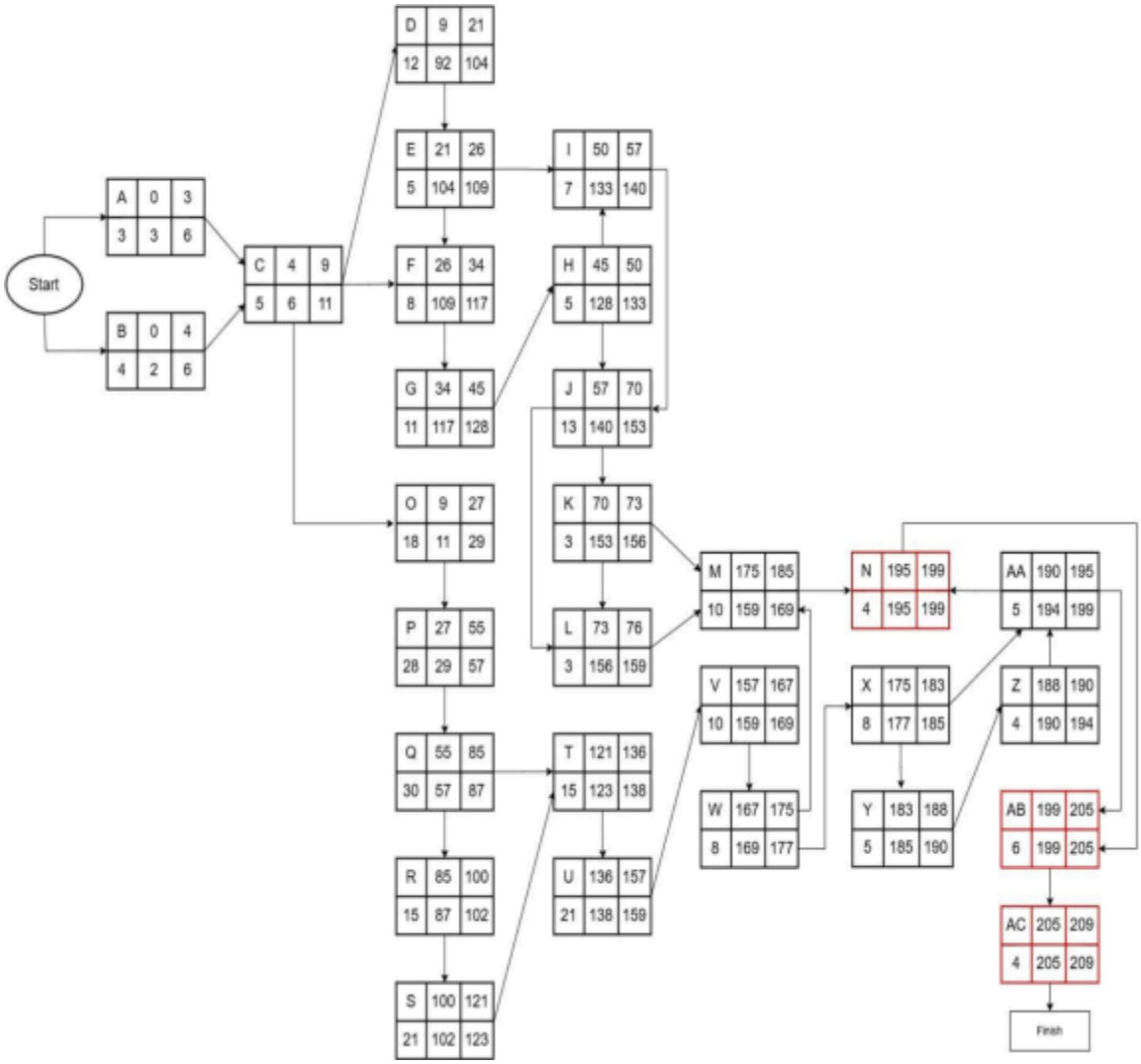
ID	ACTIVITIES	PREDECESSOR	NO. OF WEEKS
A	Group Assignment to Adviser	—	1
B	Topic Identification	—	1
C	Topic Proposal Approval	B,A	1
D	Writing Chapter 1	C	2
E	Revision – Chapter 1	D	1
F	Literature Review	C,E	1
G	Writing Chapter 2	F	2
H	Revision – Chapter 2	G	1
I	Research Method Planning	E,H	1
J	Writing Chapter 3	H,I	2
K	Revision – Chapter 3	J	1
L	Capstone 1 Proposal Defense	K,J	1
M	Revision of Chapters	L,K	2
N	Adviser Checking of Revisions	M,AA	1
O	Dataset Collection	C	3
P	Classification	O	3
Q	Model Development (Multiple Linear Regression)	P	4
R	System Design	Q	2
S	System Implementation	R	3
T	System Integration	Q,S	2
U	System Testing	T	3
V	Data Analysis	U	2
W	Writing Chapter 4	M,V	1
X	Revision – Chapter 4	W	1
Y	Writing Chapter 5	X	1
Z	Revision – Chapter 5	Y	1
AA	Final Revision	X,Z	1
AB	Final Defense Preparation	AA,N	1
AC	Final Defense	AB	1



ID	ACTIVITIES	PREDECESSOR	NO. OF WEEKS
A	Group Assignment to Adviser	—	1
B	Topic Identification	—	1
C	Topic Proposal Approval	B,A	1
D	Writing Chapter 1	C	2
E	Revision – Chapter 1	D	1
F	Literature Review	C,E	1
G	Writing Chapter 2	F	2
H	Revision – Chapter 2	G	1
I	Research Method Planning	E,H	1
J	Writing Chapter 3	H,I	2
K	Revision – Chapter 3	J	1
L	Capstone 1 Proposal Defense	K,J	1
M	Revision of Chapters	L,K	2
N	Adviser Checking of Revisions	M,AA	1
O	Dataset Collection	C	3
P	Classification	O	3
Q	Model Development (Multiple Linear Regression)	P	4
R	System Design	Q	2
S	System Implementation	R	3
T	System Integration	Q,S	2
U	System Testing	T	3
V	Data Analysis	U	2
W	Writing Chapter 4	M,V	1
X	Revision – Chapter 4	W	1
Y	Writing Chapter 5	X	1
Z	Revision – Chapter 5	Y	1
AA	Final Revision	X,Z	1
AB	Final Defense Preparation	AA,N	1
AC	Final Defense	AB	1



PERT CHART



Critical Path: N,AB, AC

Figure 2. *PERT Chart of the Proposed System*

Table 9: Activity List — Regressions

ID	Activity	Start Date	End Date	Duration (Weeks)	Predecessor
A	Group Assignment to Adviser	01/20/2026	01/22/2026	1	
B	Topic Identification	01/23/2026	01/26/2026	1	A,O
C	Topic Proposal Approval	01/27/2026	01/31/2026	1	B
D	Writing Chapter 1	02/01/2026	02/12/2026	2	C
E	Revision – Chapter 1	02/13/2026	02/17/2026	1	D
F	Literature Review	02/18/2026	02/25/2026	1	E
G	Writing Chapter 2	02/26/2026	03/08/2026	2	F
H	Revision – Chapter 2	03/09/2026	03/13/2026	1	G
I	Research Method Planning	03/14/2026	03/20/2026	1	E,H
J	Writing Chapter 3	03/21/2026	04/02/2026	2	H
K	Revision – Chapter 3	04/03/2026	04/05/2026	1	J
L	Capstone 1 Proposal Defense	04/06/2026	04/08/2026	1	E,H,K

M	Revision of Chapters 1–3	04/09/2026	04/18/2026	2	L
N	Adviser Checking of Revisions	04/19/2026	04/22/2026	1	M
O	Dataset Collection	04/23/2026	05/10/2026	3	C
P	Classification of size	05/11/2026	06/05/2026	3	O
Q	Model Development (Three Linear Regression)	06/06/2026	07/05/2026	4	P
R	System Design	07/06/2026	07/20/2026	2	Q
S	System Implementation	07/21/2026	08/10/2026	3	R
T	System Integration	08/11/2026	08/25/2026	2	S
U	System Testing	08/26/2026	09/15/2026	3	T
V	Data Analysis	09/16/2026	09/25/2026	2	U
W	Writing Chapter 4	09/26/2026	10/03/2026	1	M
X	Revision – Chapter 4	10/04/2026	10/07/2026	1	W
Y	Writing Chapter 5	10/08/2026	10/12/2026	1	X
Z	Revision – Chapter 5	10/12/2026	10/14/2026	1	Y
AA	Final Revision	10/14/2026	10/16/2026	1	X,Z
AB	Final Defense Preparation	10/17/2026	10/20/2026	1	AA
AC	Final Defense	10/21/2026	10/23/2026	1	AB

CONTEXT DIAGRAM

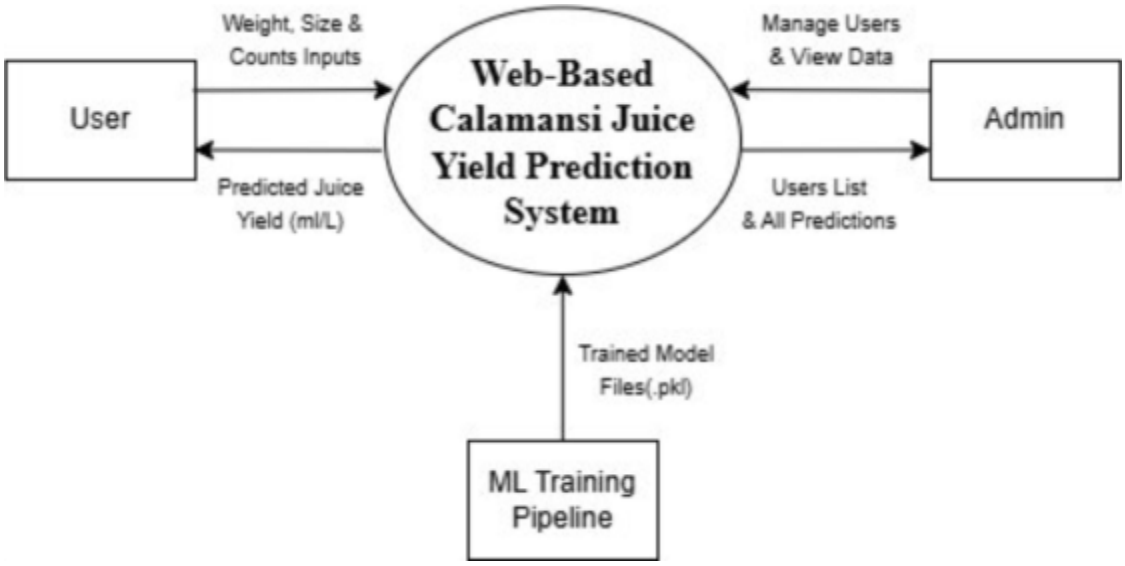


Figure 3. *Context Diagram of the Current System*

DATA FLOW DIAGRAM

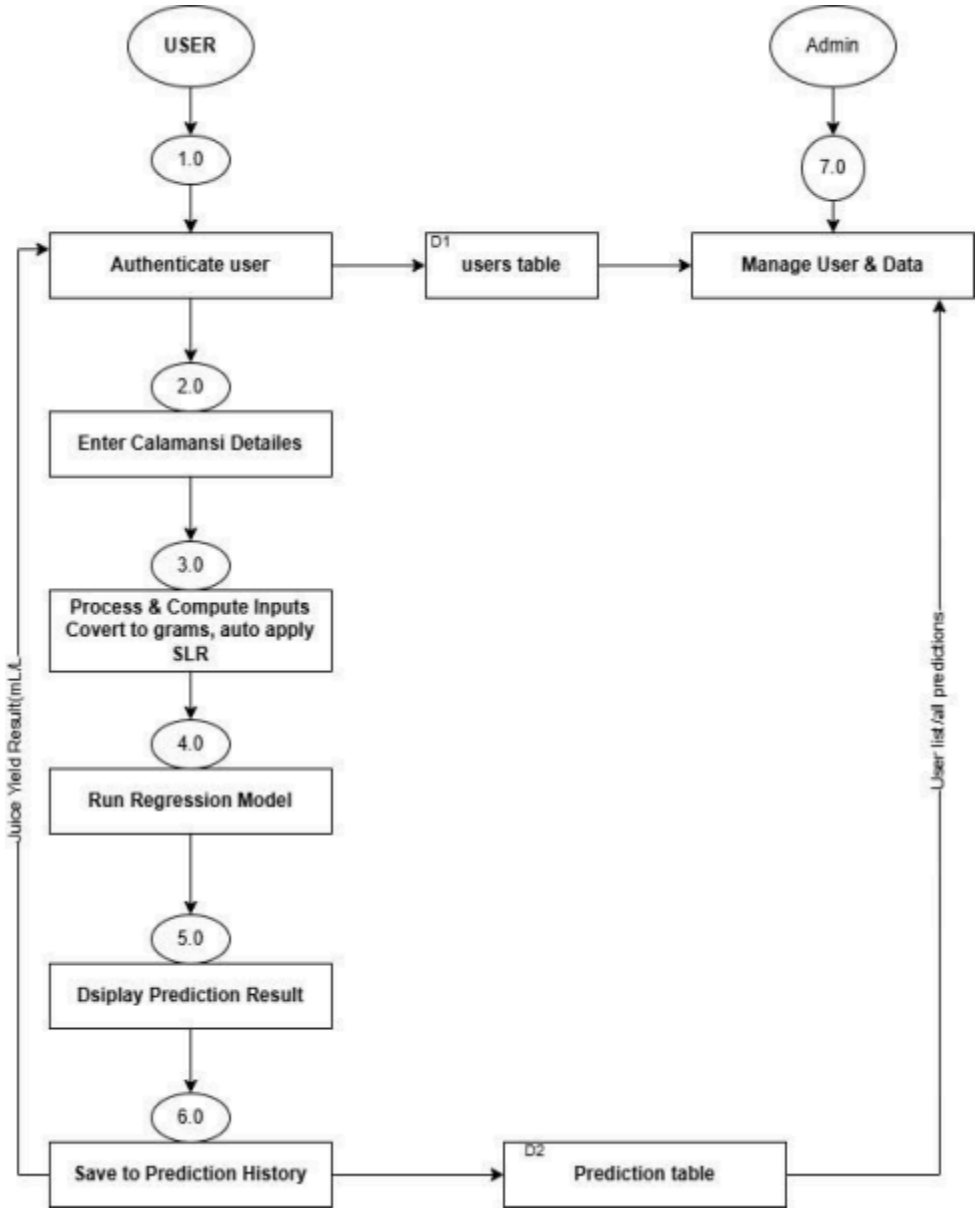


Figure 4. *Data Flow Diagram of the Study*

SYSTEM FLOWCHART

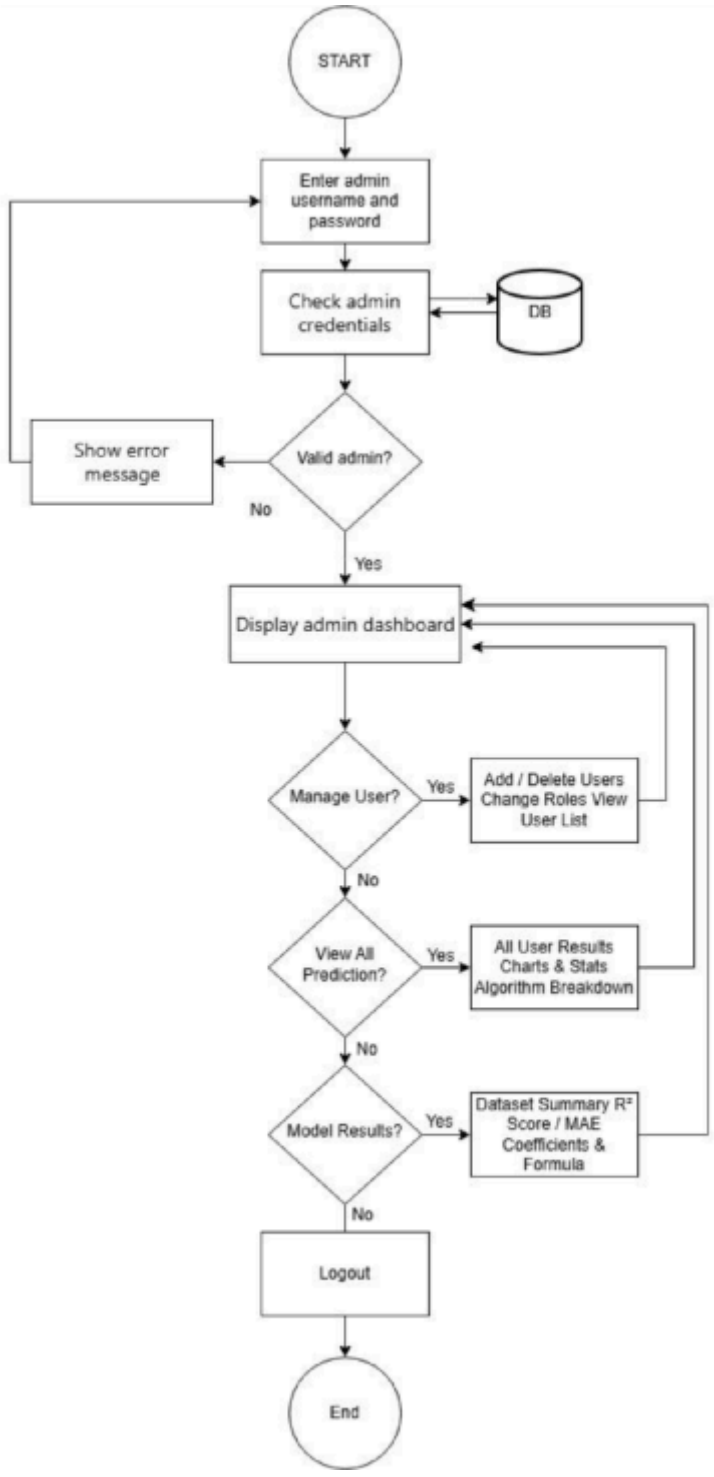


Figure 5. *System Flowchart for Admin of the Proposed System*

SYSTEM FLOWCHART

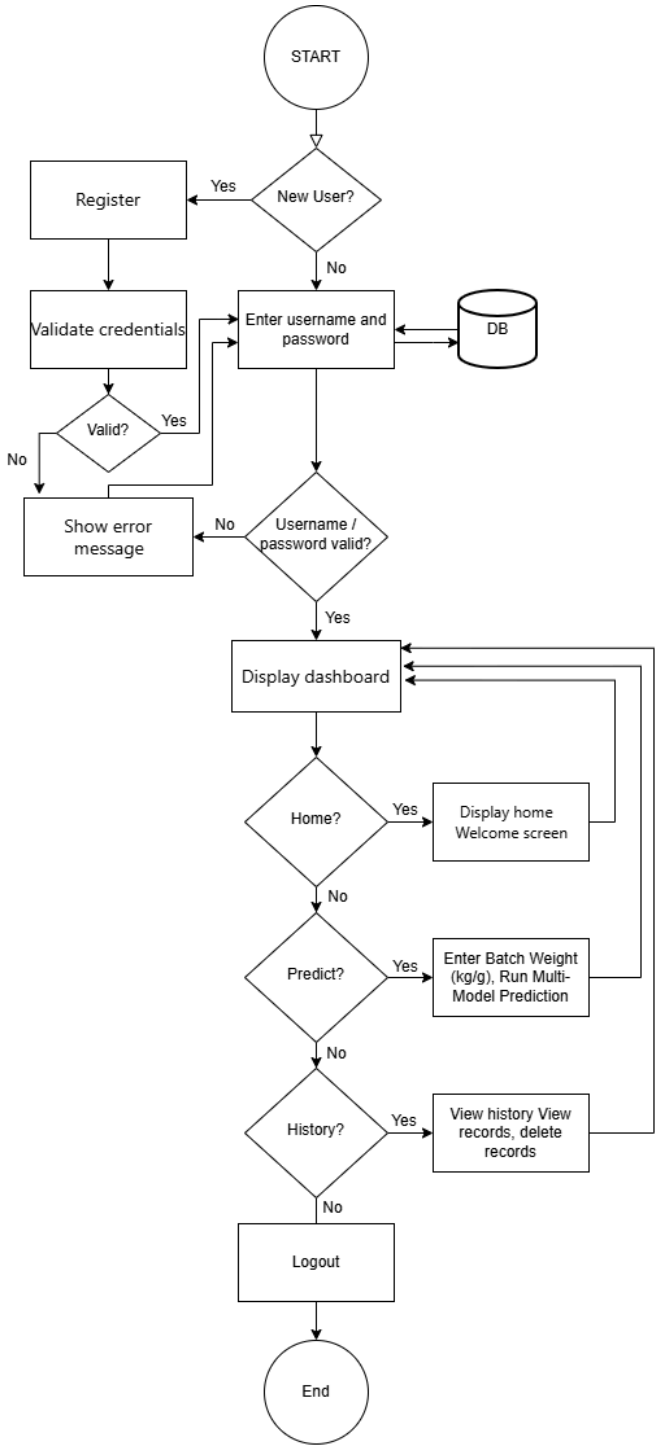


Figure 5.1 *System Flowchart for User of the Proposed System*

Entity Relationship Diagram

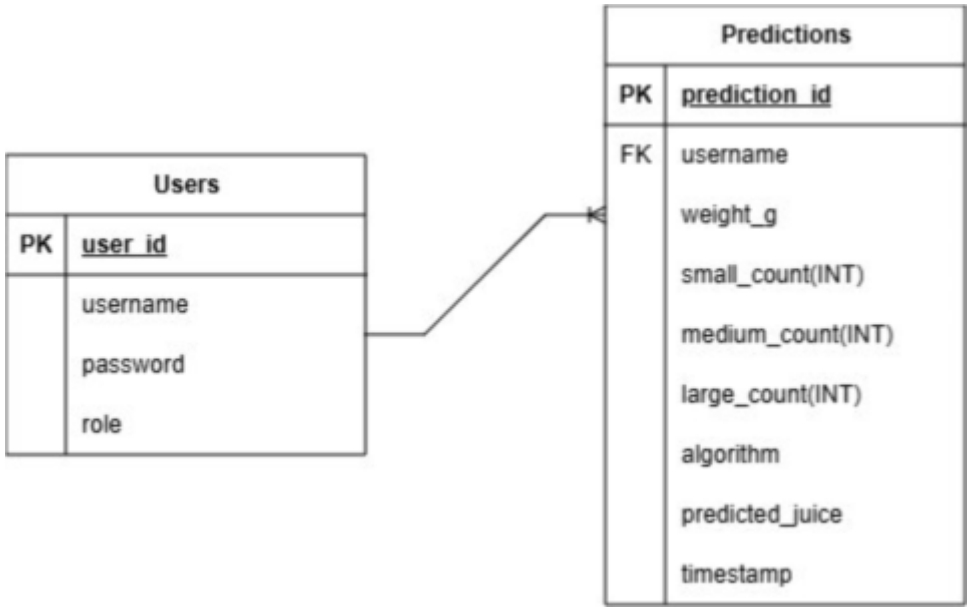


Figure 6. *ERD of the Study*

Database Normalization

The proposed Mobile Calamansi Juice Yield Prediction System uses two database tables: the Users table to manage account credentials and roles, and the Predictions table to store all prediction records submitted by users. To ensure data integrity, prevent redundancy, and optimize data operations, both tables were normalized to Third Normal Form (3NF). Normalization organizes the system data related to user credentials, roles, prediction inputs, and prediction outputs into structured tables, improving query efficiency and supporting accurate system operation and record keeping.

Normalization Process

Normal Form	Requirements	Compliance Actions
1NF	Atomic attributes; no repeating groups.	The Predictions table contains single-value attributes: id, username, weight_g, small_count, medium_count, large_count, algorithm, predicted_juice, timestamp
2NF	Full primary key dependency.	All attributes depend entirely on the primary key (id), ensuring no partial dependencies.

3NF	No transitive dependencies.	Attributes such as weight_g and algorithm are directly related to the prediction id, with no indirect dependencies.

Normalized Schema

```
CREATE TABLE Users (
  user_id BIGINT GENERATED ALWAYS AS
        IDENTITY PRIMARY KEY,
  username TEXT NOT NULL UNIQUE,
  password TEXT NOT NULL,
  role TEXT NOT NULL CHECK (role IN ('admin',
        'user'))
);
CREATE TABLE Predictions (
  id BIGINT GENERATED ALWAYS AS
        IDENTITY PRIMARY KEY,
  username TEXT NOT NULL REFERENCES
        Users(username),
  weight_g REAL NOT NULL,
  small_count INTEGER NOT NULL,
  medium_count INTEGER NOT NULL,
  large_count INTEGER NOT NULL,
  algorithm TEXT NOT NULL,
  predicted_juice REAL NOT NULL,
  timestamp TIMESTAMPTZ NOT NULL
        DEFAULT now()
);
```

Verification and Benefits

Tested: Sample data was inserted and queried successfully. Normalization ensures that updates to individual prediction records remain consistent and do not introduce redundancy across user accounts.

Query Efficiency: Retrieving records for evaluation is fast and reliable using simple queries such as: SELECT * FROM Predictions WHERE algorithm = 'Polynomial Regression (d=2)';

Data Integrity: Each prediction has a unique ID, preventing duplication and ensuring consistent records across all three algorithm logs.

Scalability: The normalized table can handle growth as more predictions are added by users over time without redundancy or performance issues.

Discussion: Normalizing the Predictions table ensures that all user prediction records remain clean, consistent, and free of redundant data. This is critical for a system that logs predictions from three different algorithms, as it allows administrators to query and analyze algorithm usage patterns efficiently while maintaining fast data retrieval.

Use Case

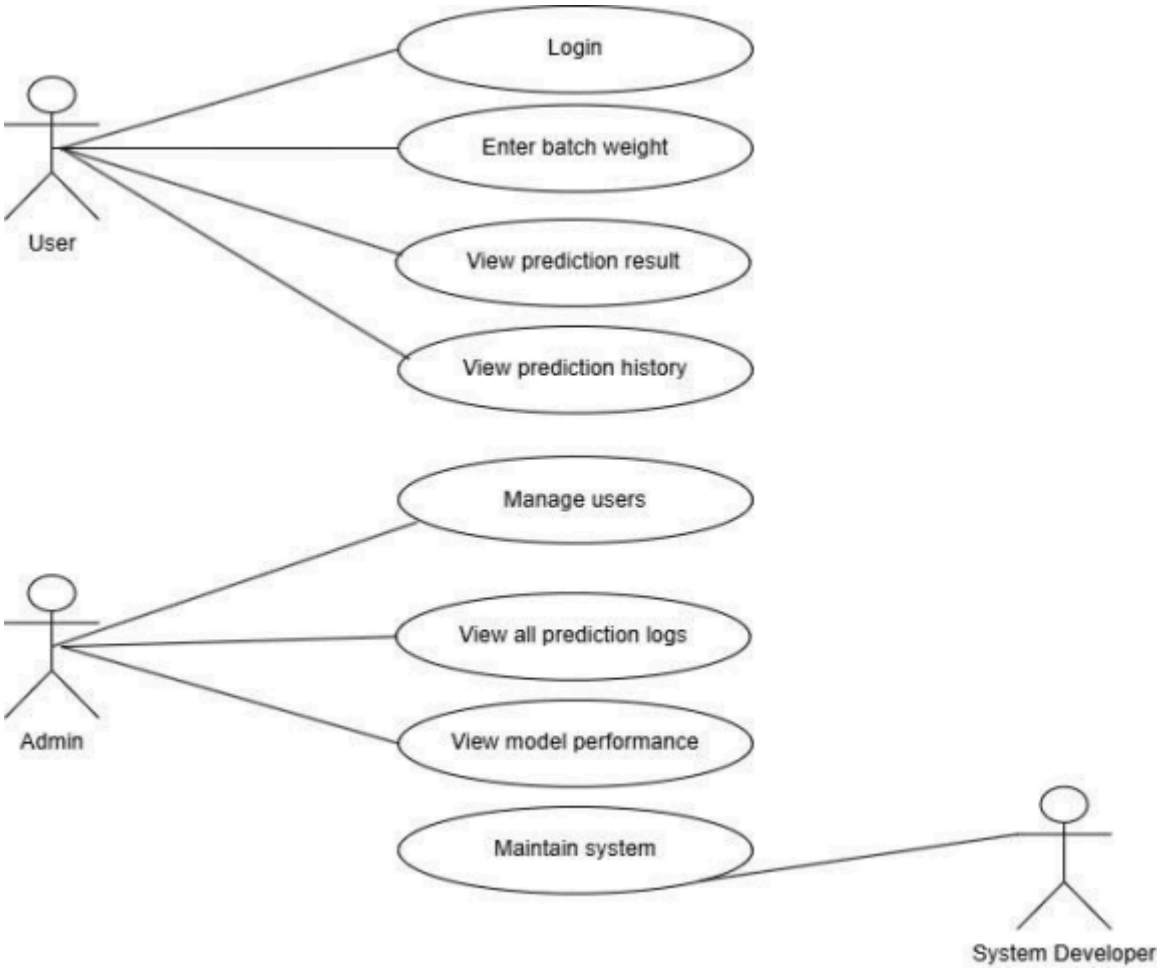


Figure 7. *Use Case of the Study*

The regular-user workflow begins when the user logs in and enters the total calamansi batch weight. The user selects kilograms or grams, and the system converts the value to grams before running the prediction engine. The predicted juice yield is displayed in milliliters and liters for all three models and saved to the user's prediction history. Regular users are not required to enter tedious size counts in the field.

Sorting by size was performed by the researchers only during data collection to build the training dataset. In actual use, the juice processor is not required to sort the calamansi. The user enters only the total batch weight, and the system applies the representative unit weight derived from the research dataset to evaluate the Multiple Linear and Polynomial Regression models, which use both weight and size.

4.3 Description of the Prototype

The prototype developed in this study is a Calamansi Juice Yield Prediction Mobile Application built using Flutter and Dart. The system trains and compares three regression models to estimate juice yield based on two observable physical characteristics of the calamansi: weight and size category.

The main objective of the prototype is to provide a non-destructive and automated method for predicting calamansi juice yield without manually squeezing every calamansi. The prototype is composed of hardware and software components that work together to perform input processing, prediction modeling, and result display.

1. **Input Module –**The regular-user input form accepts the total calamansi batch weight and its unit of measurement. Users can select kilograms or grams. The system validates the weight, converts it to grams, and automatically applies all three regression models (Simple Linear, Multiple Linear, and Polynomial

2. Data Preprocessing and Feature Encoding – Each submitted input is preprocessed before being passed to the regression models. The total weight entered in kilograms is converted to grams. Because users are not required to sort calamansi by size, the system uses the representative unit weight derived from the research dataset (approximately 12.65 g, the mean fruit weight) as the basis for the size feature. Simple Linear Regression uses weight only, while Multiple Linear Regression and Polynomial Regression use both weight and size. This encoding step replicates the preprocessing applied during model training, ensuring consistency between training and inference.

3. Machine Learning Model – Three regression models trained on the collected calamansi samples are used. (a) Simple Linear Regression uses only weight as a predictor. It achieved $R^2 = 0.7099$ and MAE = 0.5294 ml. (b) Multiple Linear Regression uses weight and size as predictors. It achieved $R^2 = 0.7100$ and MAE = 0.5292 ml. (c) Polynomial Regression (degree 2) uses weight, size, weight², weight×size, and size² as predictors. It achieved the best performance with $R^2 = 0.7102$ and MAE = 0.5279 ml. All three models are stored as separate .pkl files and loaded at runtime for real-time predictions.

4. System Interface – The mobile application delivers an intuitive, touch-optimized user interface on Android devices. Predictions are computed instantaneously upon entering batch weights in kilograms or grams, displaying comparative results across Simple Linear, Multiple Linear, and Polynomial regression models alongside dedicated history and admin dashboards.

Project Title

Development of a Mobile Calamansi Juice Yield Prediction System Using Regressions

Project Description

This project develops a mobile application that predicts the juice yield of calamansi based on observable physical characteristics specifically weight and size category by evaluating and comparing three linear regression algorithms: Simple Linear Regression, Multiple Linear Regression, and Polynomial Regression (degree 2). The system provides real-time predictions through a Flutter mobile application accessible on handheld Android devices.

Objectives:

1. To collect and categorize calamansi samples into three size groups (small, medium, and large) based on defined physical criteria, and to record the individual weight (in grams) of each sample using a calibrated digital kitchen scale, ensuring a complete dataset with zero missing values during the data gathering phase.
2. To measure the actual juice yield (in milliliters) of each categorized calamansi sample using a standardized manual extraction method and calibrated syringe to establish the ground truth dataset during the experimentation phase.
3. To train and compare three linear regression models—Simple Linear Regression (weight only), Multiple Linear Regression (weight and size), and

Polynomial Regression of degree 2 (weight, size, and their squared and interaction terms)—using an 80:20 training-to-test split, targeting an R^2 score of at least 0.80 and the lowest achievable Mean Absolute Error (MAE) for the best-performing model during the model development phase.

4. To develop a mobile application using Flutter and Dart that accepts the total calamansi batch weight in kilograms or grams, displays the predicted juice yield from the three regression models, and records prediction history, with an administrator module for reviewing user logs and model performance metrics.

5. To evaluate the performance of the mobile application based on prediction response time (target: under 2 seconds per prediction) and the accuracy of its predictions using R^2, MAE, and RMSE.

4.4. Table 10. Implementation Plan Table

Module	Specific Objective/s	Hardware Requirements	Software Requirements	Duration (Day)	Person Involved	Estimated Budget (PHP)	Risk Assessment and Mitigation Strategies	Evaluation and Monitoring
Data Collection	Collect and sort calamansi samples by size and weight.	Digital kitchen scale, standard syringe	Microsoft Excel	10 Days	Researcher	2000	Inconsistent size assessment; mitigate by defining clear size categories small ≤10 g, medium 10.1–14 g, large >14 g) before collection begins.	Inspect dataset for completeness and verify no missing values before model training.
Juice Yield Measurement	Measure actual juice yield (mL) of each sample using a hand-press juicer and standar	hand-press juicer Standard syringe	None	10 Days	Researcher	300	Measurement inaccuracies; mitigate by using standardized extraction methods and repeated	Compare recorded juice yield values and verify measurement consistency across samples.

	d syringe to serve as ground truth						measurements.	
Data Preprocessing and Model Development	Encode size as numeric, split dataset 80:20, and train and compare all three regression models achieving $R^2 \geq 0.80$.	Laptop / Personal Computer (min. 8GB RAM)	Python, scikit-learn, pandas, NumPy, Jupyter Notebook	20 Days	Researcher / Developer	0	Model underfitting; mitigate by reviewing feature selection and trying polynomial features if linear models underperform.	Evaluate all three models using R^2 and MAE on the test set and compare against target metrics.

Mobile Application Deployment	Develop a Flutter mobile application for real-time calamansi juice yield prediction.	Laptop / Personal Computer, stable internet connection	Flutter SDK, Dart, VS Code	5 Days	Researcher / Developer	0	System errors or incorrect output display; mitigate by debugging and testing the application before deployment.	Conduct system testing and verify that all prediction outputs are correct and displayed properly.

Ethical Considerations

Declaration of No Conflict of Interest

The researchers affirm that this study was developed purely for academic purposes as part of the requirements for the Bachelor of Science in Information Technology program at Leyte Normal University. No financial assistance, sponsorship, or external influence from any company, organization, or individual was involved in the design, development, or evaluation of the system.

The study was carried out independently to ensure that the results remain objective and unbiased. The researchers also confirm that there are no personal, professional, or financial conflicts of interest that could affect the integrity of the research.

The dataset used to train the regression models was built from calamansi samples sourced from local market vendors and suppliers in Tacloban City. Only the physical measurements of the samples, namely size category, weight, and extracted juice yield, were recorded. No names, contact details, addresses, or other private identifiers of the vendors were gathered or stored. The study did not involve interviews or any form of data collection that requires sensitive personal information.

Human participation was limited to the testing of the developed mobile application. An IT evaluator took part in the alpha testing, a beta tester took part in the beta testing, and a juice-business respondent answered the User Acceptance Testing (UAT) questionnaire. These participants took part voluntarily, were informed of the purpose of the study, and are identified in the manuscript only by their roles. They were not required to disclose personal or confidential information, and their responses were used only to evaluate the system.

The only user data handled by the application are the username and password of each account and the batch weights and predicted yields saved in the prediction history. No sensitive personal information beyond the username is collected. Because of this, the risk of privacy issues, identity exposure, or data misuse is very minimal. The researchers ensured that all procedures remain ethical, transparent, and centered on academic integrity and data privacy.

Confidentiality and Data Protection Plan (Data Privacy Act Compliant)

This study fully observes the Data Privacy Act of 2012 (Republic Act No. 10173) to ensure that all data used in the research are properly protected, handled responsibly, and used only for legitimate academic purposes.

The dataset and the model training process involve only non-personal agricultural data, specifically measurements of calamansi weight, size, and juice yield. The application stores only the data needed for its operation, which are the username, the hashed password, the role of the account, and the prediction records with timestamps linked to that account. Passwords are secured using SHA-256 hashing before they are stored in the database and are never saved in plain text. Role-based access control allows regular users to view only their own prediction history, while the administrator can access all prediction logs, user management tools, and model performance results. Data are stored in a Supabase (PostgreSQL) cloud database, and data in transit between the mobile application and the database are protected by TLS/HTTPS encrypted connections. The trained regression models and the dataset are kept in protected server-side directories that cannot be accessed through the mobile application interface. In cases where unintended personal data appear during data collection or testing, these are immediately removed or anonymized before storage to ensure that privacy is not compromised at any point.

All datasets, trained models, system logs, completed testing forms, and evaluation outputs are securely stored in password-protected devices accessible only to the researchers. These files are used strictly for academic development, testing, and evaluation of the system and are not shared publicly without proper authorization or

ethical approval. The researchers also apply data minimization principles by collecting only the information necessary for system training and evaluation. No unnecessary or unrelated data are retained. After the completion of the study, temporary files, test accounts, and unused data will be properly deleted or archived following institutional research guidelines.

For any user interaction or system testing, participants remain anonymous and are reported only by role. They are not required to provide personal information beyond what is needed to operate the system. All generated results and user interactions are treated with strict confidentiality. Through these measures, the study ensures that all data are handled with care, integrity, and responsibility while fully complying with ethical standards and legal requirements under the Data Privacy Act of 2012.

Dissemination Plan

The findings of this study will be formally presented to the faculty panel of Leyte Normal University as part of the completion requirements for the Bachelor of Science in Information Technology program. A copy of the research output will also be submitted to the university's research office and library for academic reference and future studies.

In addition, the developed mobile calamansi juice yield prediction system will be demonstrated to selected users such as juice processors. This demonstration aims to show how the system can provide practical value by helping users estimate juice yield efficiently without the need for manual extraction.

4.5 Implementation Results and Discussion

This section presents the results of the implementation of the Mobile Calamansi Juice Yield Prediction System. It shows the actual screens of the developed mobile application for both user and administrator roles, followed by the discussion of the regression results displayed by the system. The application was developed using Flutter, with Supabase for authentication and prediction history, and Python-trained regression models for the yield estimates.

4.5.1 User Authentication Module

Figure 8 shows the sign-in screen, which is the first screen displayed when the application is opened. It contains the system title, the name of the institution, and the Sign In and Create Account tabs. The user enters a registered username and password, with an option to show or hide the password, then taps the Sign In to Predictor button. Credentials are verified through Supabase before access to the predictor is granted.

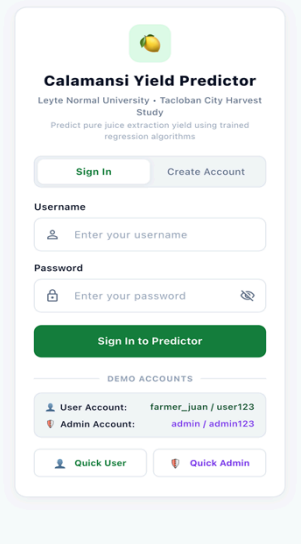


Figure 8. *Sign-in Screen of the Calamansi Yield Predictor*

Figure 9 shows the Create Account screen for new users. The user chooses a username, creates a password of at least four characters, and confirms the password before tapping the Create Account & Sign In button. The system registers the account with the standard user role and signs the user in right away.

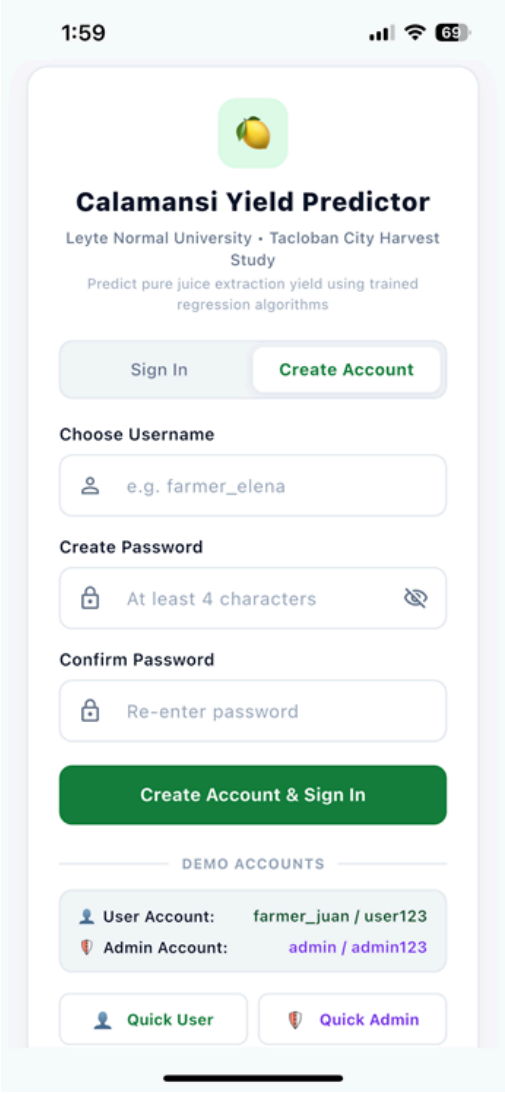


Figure 9. *Create Account Screen*

4.5.2 Yield Prediction Module

Figure 10 shows the main screen of the system, labeled Estimate Yield. The header displays the role badge and the username of the signed-in user, together with the history and Sign Out buttons. The user enters the total calamansi batch weight, chooses the unit (kg or g), and may use the Quick Select buttons for common weights. Tapping Run all 3 models starts the prediction. Before any run, the result area displays a Ready to predict message.

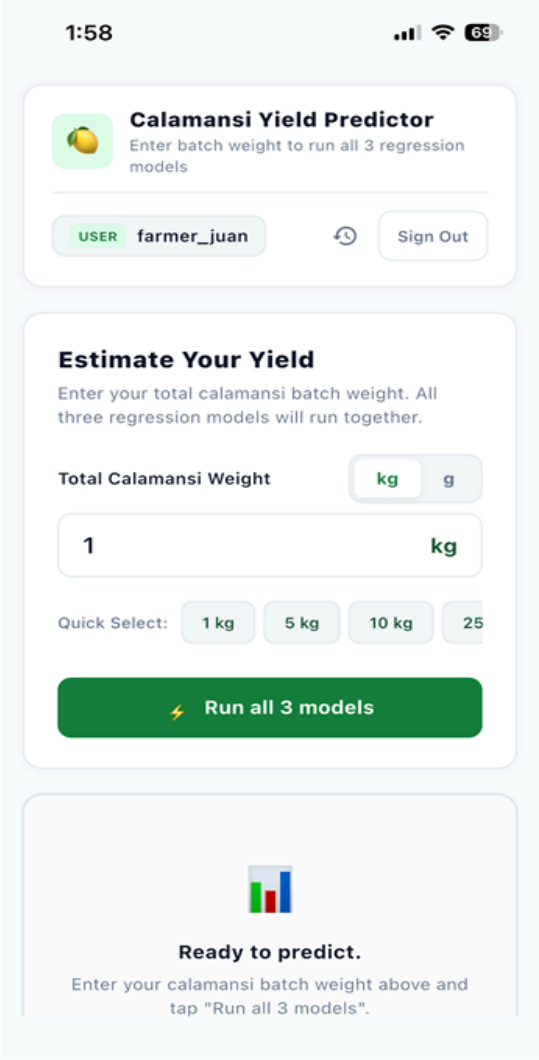


Figure 10. *Main Yield Predictor Screen*

Figure 11 shows the screen after a batch weight of 15 kg was entered and the models were run. The Prediction Results section displays a badge showing the general harvest average (mixed sizes) and the estimated number of calamansi in the batch.

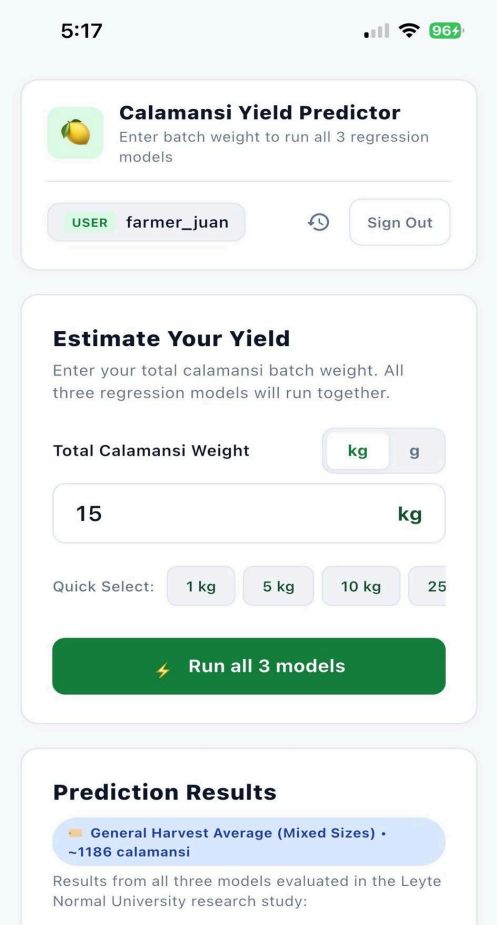


Figure 11. *Batch Weight Input and Prediction Results Header*

Figure 12 shows the result cards of the Simple Linear Regression and the Multiple Linear Regression models. For the 15 kg input, Simple Linear Regression, which uses weight only, predicted 6132.55 ml (about 6.1326 L), while Multiple Linear Regression, which uses weight and size, predicted 6132. 66 ml (about 6.1327 L). Each card also lists the model inputs and its evaluation metrics, namely R^2, test MAE, RMSE, MSE, and MAPE.

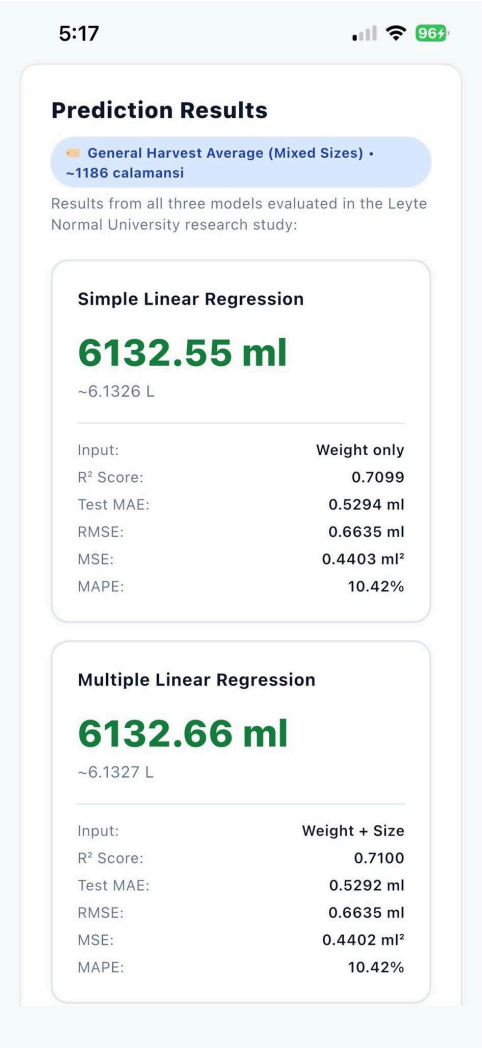


Figure 12. *Simple and Multiple Linear Regression Results*

Figure 13 shows the result card of the Polynomial Regression (d=2) model, which is marked with a Best Model label. It predicted 6151.93 ml (about 6.1519 L) using weight, size, and their squared and interaction terms as inputs. A recommendation box below the card explains why this model is suggested: it obtained the highest coefficient of determination (R^2 = 0.7102) and the lowest test error (MAE = 0.5279 ml) among the three models on the unseen test samples, which shows that it better captures the non-linear variation in calamansi extraction.

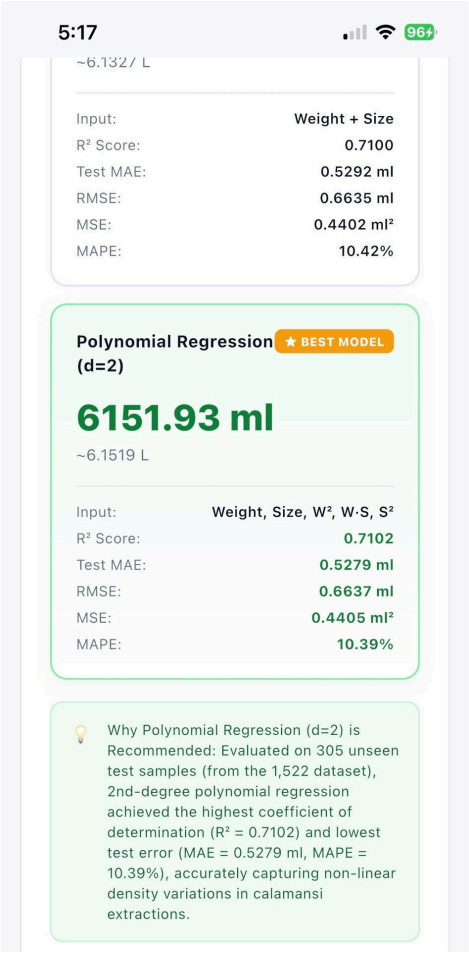


Figure 13. *Polynomial Regression Result and Model Recommendation*

Figure 14 shows the Your Recent Harvests section, which lists the batch extractions and yield estimates recorded in Supabase for the signed-in user. Each row shows the date and time, the batch weight, and the assigned grade. This allows farmers to review earlier estimates and compare yields across harvests without recomputing them.



Figure 14. *Your Recent Harvests Table*

4.5.3 Administrator Module

Figure 15 shows the User Management tab of the Admin Research & Governance Console. It lists the authorized farmer and system researcher accounts with their role, status, and date joined. The administrator can create a new user, suspend or reactivate an account, promote a user to administrator, reset a password, or delete an account. The signed-in administrator account is marked and protected from these actions, which prevents accidental loss of administrative access.

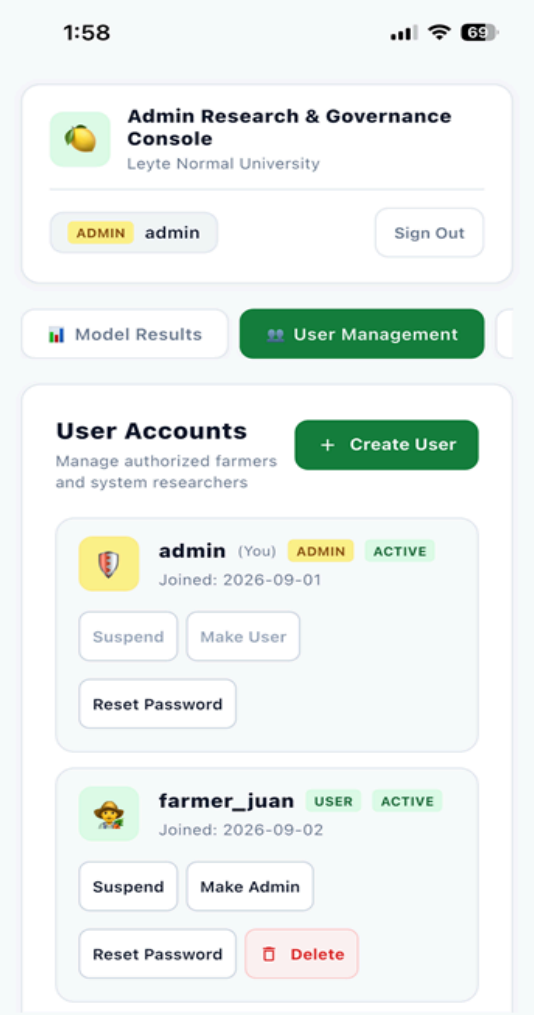


Figure 15. *Admin User Management Screen*

Figure 16 shows the Prediction Logs tab, labeled User Prediction History. It records the batch weights and outputs run on the system, with columns for date and time, user, and batch weight. A Clear Logs button allows the administrator to remove the stored history.

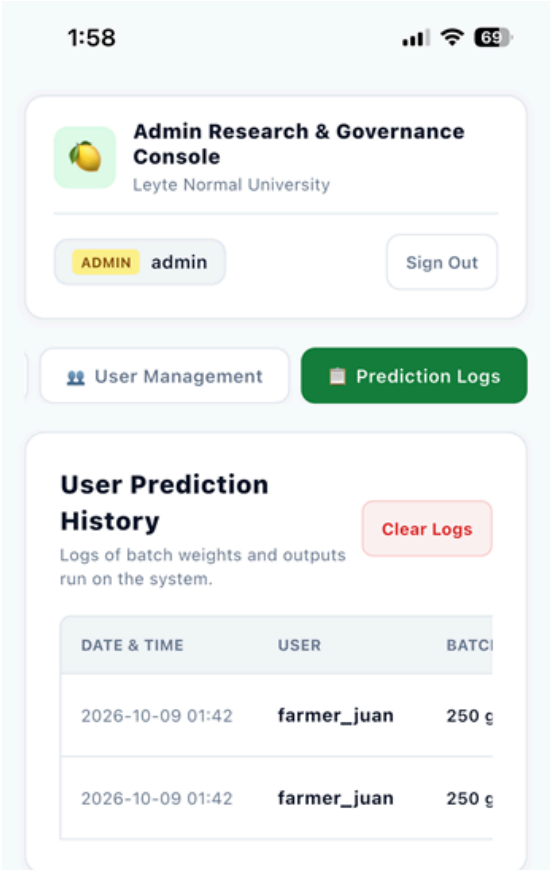


Figure 16. *Admin Prediction Logs Screen*

Figure 17 shows the next columns of the prediction logs, which are the batch weight and the assigned grade. In the recorded runs, a 250 g batch was assigned the General Harvest Average (Mixed Sizes) grade.

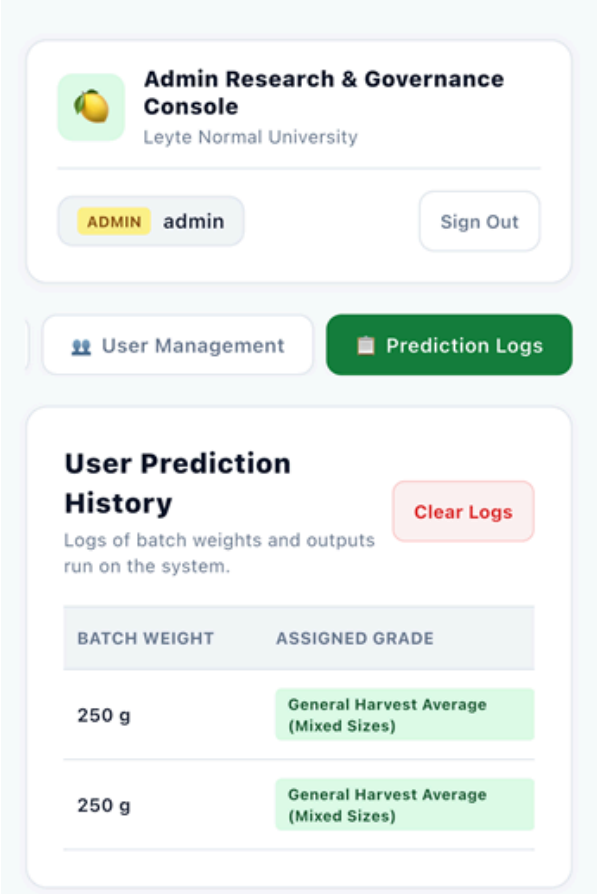


Figure 17. *Prediction Logs: Batch Weight and Assigned Grade*

Figure 18 shows the model output columns of the logs for the Simple Linear, Multiple Linear, and Polynomial models. For the recorded 250 g batch, the models returned 102.03 ml, 102.03 ml, and 102.35 ml, respectively.

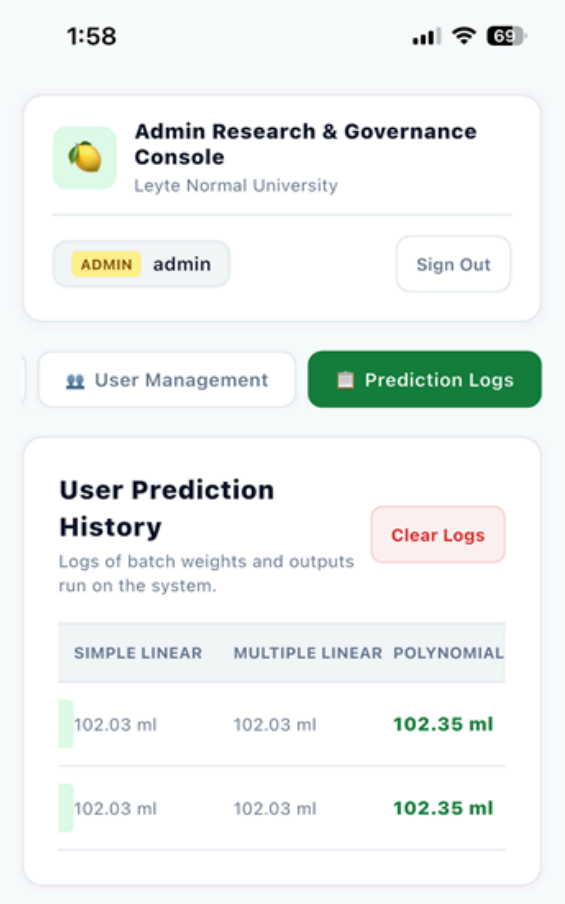


Figure 18. *Prediction Logs: Model Outputs*

Figure 19 shows the right-most part of the prediction logs, where the outputs of the Multiple Linear and Polynomial (Best) models are displayed side by side. The Polynomial output is highlighted in bold green, which makes it easy to identify the best-performing model in each logged run.

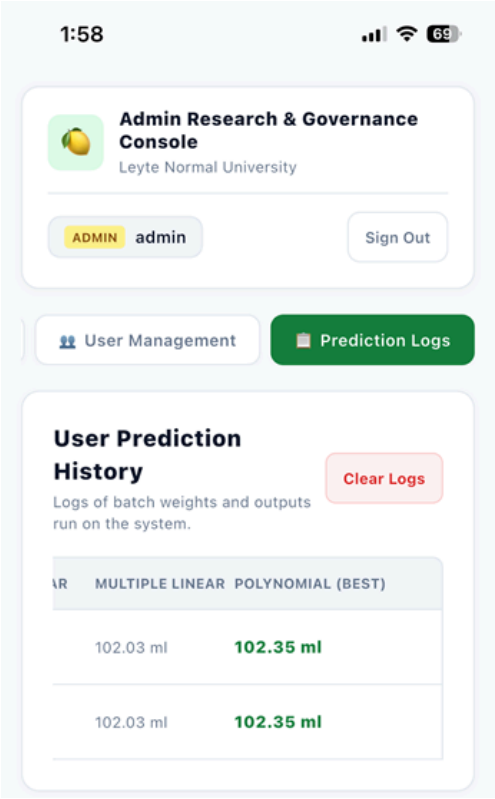


Figure 19. *Prediction Logs: Multiple Linear and Polynomial (Best) Outputs*

Figure 20 shows the Model Results tab, starting with the Dataset Summary of the Tacloban City harvest study. The dataset has 1,522 physical calamansi extractions, divided into 1,217 samples (80%) for training and 305 samples (20%) for testing. The Size Distribution table classifies the samples as Small (Size 1) for weights of 10.0 g or less, Medium (Size 2) for weights from 10.1 g to 14.0 g, and Large (Size 3) for weights above 14.0 g.

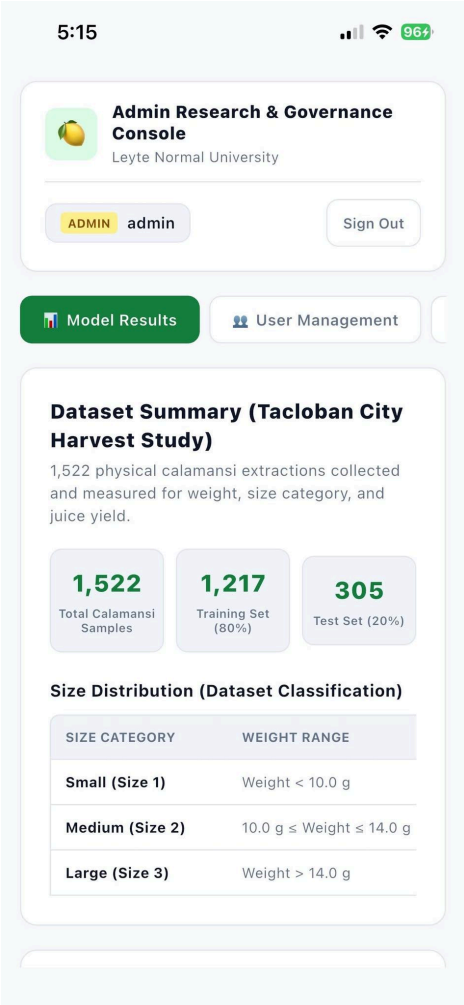


Figure 20. *Model Results: Dataset Summary*

Figure 21 shows the Algorithm Performance Comparison of the three regression models tested on unseen test samples (N=305), together with the trained mathematical formulas. The formulas are Juice (ml) = 0.4569 × Weight − 0.6080 for Simple Linear Regression; Juice (ml) = 0.4580 × Weight − 0.0053 × Size − 0.6108 for Multiple Linear Regression; and Juice (ml) = −0.5480 + 0.4357(W) + 0.0865(S) − 0.0031(W²) + 0.0474(W·S) − 0.1641(S²) for Polynomial Regression (Degree 2). The Polynomial model is highlighted as the optimal fit because it has the lowest test error.

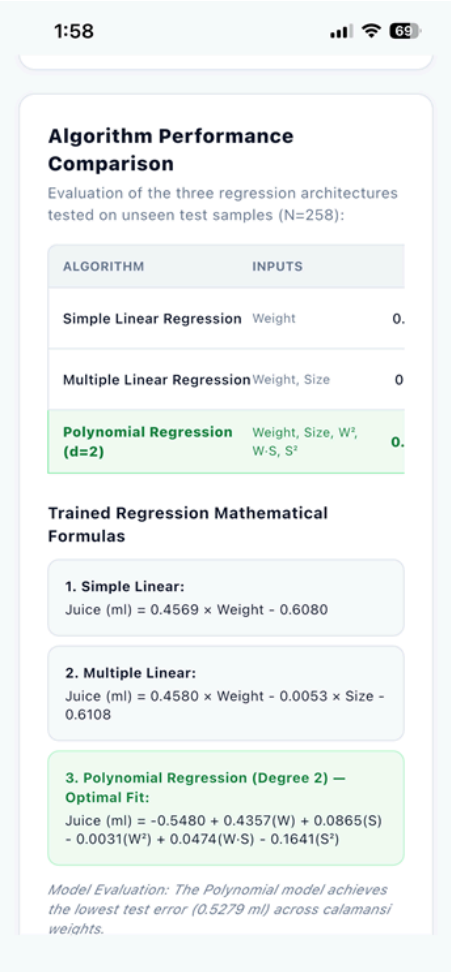


Figure 21. *Algorithm Performance Comparison and Trained Formulas*

Figure 22 shows the Algorithm Performance Comparison table in the Model Results tab, evaluated on the unseen test samples (N=305). The table lists the R^2, MAE, RMSE, MSE, and MAPE of the three models. Simple Linear Regression obtained an R^2 of 0.7099, MAE of 0.5294 ml, RMSE of 0.6635 ml, MSE of 0.4403 ml^2, and MAPE of 10.42%. Multiple Linear Regression obtained 0.7100, 0.5292 ml, 0.6635 ml, 0.4402 ml^2, and 10.42%. Polynomial Regression (Degree 2), highlighted in green, obtained the best values at 0.7102, 0.5279 ml, 0.6637 ml, 0.4405 ml^2, and 10.39%.

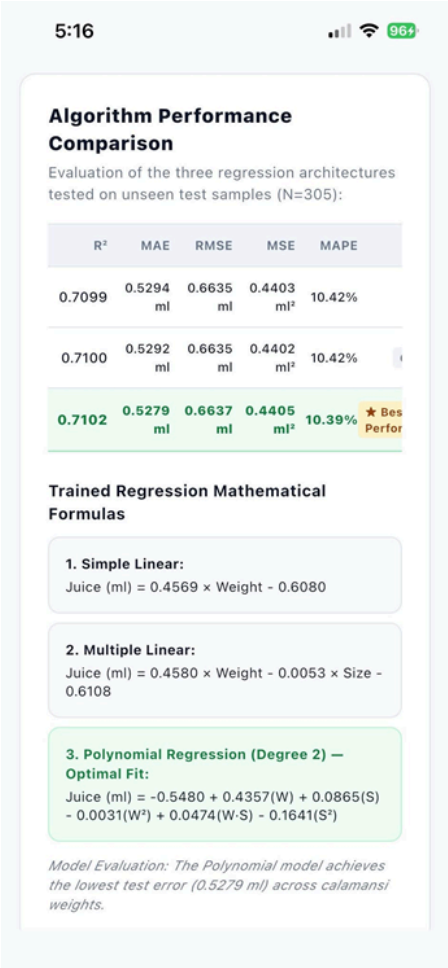


Figure 22. *Algorithm Performance Comparison: Evaluation Metrics*

Figure 23 shows the right side of the same table, which has an Outcome column that labels Simple Linear Regression as the Baseline, Multiple Linear Regression as the Comparative model, and Polynomial Regression as the Best Performing model. Below the table are the trained regression formulas used by the system, followed by a model evaluation note stating that the Polynomial model achieves the lowest test error (0.5279 ml) across calamansi weights.

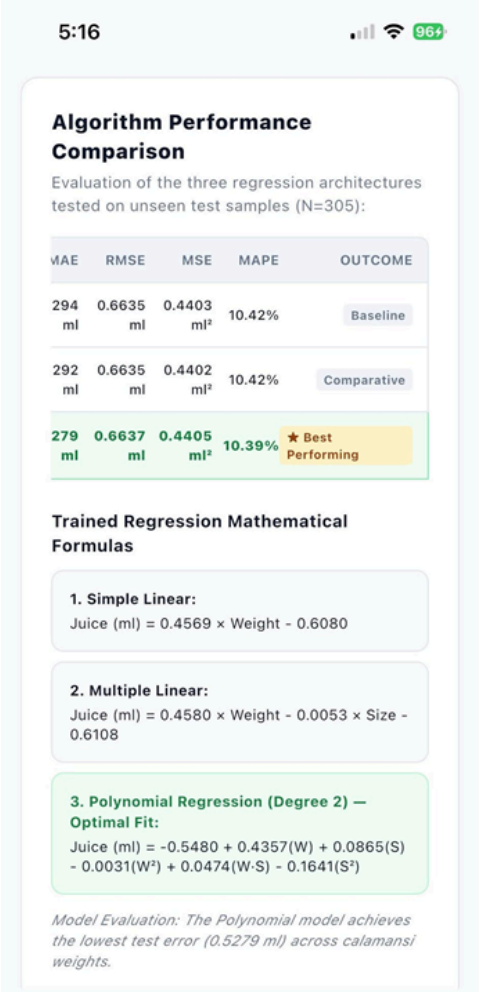


Figure 23. *Algorithm Performance Comparison: Model Outcome and Trained Formulas*

4.5.4 Black-Box Testing Results

Black-box functional testing of the Calamansi Yield Predictor mobile application was carried out in two rounds, followed by a user acceptance survey. In the alpha round, an IT evaluator (a Computer Programmer II) executed 12 functional test cases on version 1.0.0 of the application and marked each as Pass, Fail, or N/A. In the beta round, a tester executed 14 test actions while the researcher recorded the results as Pass or Fail. A juice-business respondent then rated 13 User Acceptance Testing (UAT) statements on a five-point scale (5 = Strongly Agree, 4 = Agree, 3 = Neutral, 2 = Disagree, 1 = Strongly Disagree) and answered three open-ended questions. In keeping with the confidentiality plan of the study, participants are identified only by role.

4.5.4.1 Alpha Testing

The alpha test cases evaluate the externally visible behavior of the application, including login, unit conversion, batch weight input, the output of the three regression models, the display of the best-performing model, input validation, offline use, and repeated predictions, without inspecting the source code. The evaluator performed each action, recorded what actually happened, and marked Pass only if the observed result matched the expected result.Conducted by an external IT expert (Computer Programmer II).

Table 11. Alpha Black-Box Test Cases and Results

ID	Test Case	Test Action / Input	Expected Result	Result
BB-01	Launch application	Open Calamansi Yield Predictor from the device launcher.	The application opens to its login / main screen without crashing.	Pass

BB-02	User login with valid credentials	Enter valid credentials and tap Sign In.	The application authenticates successfully and transitions to the main yield predictor screen.	Pass
BB-03	Authentication rejection on invalid credentials	Enter an invalid username or incorrect password and attempt to sign in.	The app denies access, displays an error message, and does not open unauthorized views.	Pass
BB-04	Unit toggle switching (kg ↔ g)	Enter 2.5 in kg mode, tap the g button, then tap kg again.	The input value dynamically converts (kg × 1000 = g; g ÷ 1000 = kg) without clearing input.	Pass
BB-05	Preset batch weight selection	Tap any preset harvest weight chip button (e.g., 1 kg, 5 kg, 10 kg, or 25 kg).	The weight field auto-populates with the selected value and automatically triggers prediction.	Pass
BB-06	Submit valid harvest batch weight	Enter a valid harvest weight (e.g., 10.0 kg) and tap "Run all 3 models".	The application accepts the input and presents the prediction results on screen without freezing.	Pass
BB-07	Check 3-model parallel prediction outputs	Inspect the three algorithm result cards displayed on screen.	The app displays predicted juice yields in both milliliters (ml) and liters (L) for SLR, MLR, and Polynomial models.	Pass
BB-08	Check best-performing model identification	Inspect the visual styling and badge labeling across the three algorithm result cards.	Polynomial Regression is highlighted with a green border and tagged with "BEST PERFORMING MODEL."	Pass
BB-09	Check harvest average and fruit count summary	Inspect the classification badge displayed above the prediction results.	Badge displays General Harvest Average (Mixed Sizes) along with estimated fruit count (e.g., ~803 calamansi for 10 kg).	Pass
BB-10	Submit invalid / non-positive weight input	Enter zero (0), a negative number (-5), letters (abc), or leave empty and tap Run.	The app blocks execution, shows "Please enter a valid weight greater than zero," and does not crash.	Pass
BB-11	Offline prediction and local persistence	Turn off device Wi-Fi and mobile data, then execute a harvest batch prediction.	The app calculates yield normally and saves the record to local storage without throwing network errors.	Pass

BB-12	Repeat prediction under the same session	Run a prediction for 1 kg, then immediately enter 5 kg and run again under the same session.	Both runs complete without crashing; the second run replaces the display cleanly, and both logs appear in history.	Pass

Note. Evaluator: Computer Programmer II. Application version: v1.0.0 (Production Build). Models: Simple Linear Regression (SLR), Multiple Linear Regression (MLR), and Polynomial Regression (degree 2).

As shown in Table 11, all 12 test cases passed. The application launched without crashing (BB-01), accepted valid credentials and rejected invalid ones (BB-02, BB-03), and converted between kilograms and grams without clearing the input (BB-04). It accepted both preset and manually entered batch weights (BB-05, BB-06), displayed the predicted juice yield in milliliters and liters for the three models, and highlighted Polynomial Regression as the best-performing model (BB-07, BB-08). It also showed the harvest average and estimated fruit count (BB-09), blocked zero, negative, non-numeric, and empty inputs with an error message (BB-10), calculated and saved a prediction without an internet connection (BB-11), and handled repeated predictions in the same session with both records appearing in history (BB-12).

Table 12. Summary of Alpha Testing Results

Measure	Result
Total functional test cases	12
Passed	12
Failed	0
N/A / not executed	0
Pass rate	100%
Evaluator recommendation	Acceptable for tested functions

Note. The evaluator's recommendation options were Acceptable for tested functions, Needs improvement, and Retest recommended.

All 12 test cases passed and the evaluator recommended the application as acceptable for the functions tested. The evaluator's form also states that the application is an experimental decision-support tool and that its juice yield outputs are mathematical estimates derived from the regression models, and that actual commercial juice extraction may vary depending on fruit maturity, peel thickness, ambient moisture, and mechanical press efficiency.

4.5.4.2 Beta Testing

In the beta round, the tester performed each action in the application, and the researcher recorded what the application actually did and checked Pass or Fail against the expected result. Practical field and user acceptance testing conducted with a juice-business partner and facilitated by the researchers .The 14 test actions were grouped into four categories, shown in Table 13.

Table 13. Beta Black-Box Test Results

No.	Test Action / Input	Expected Result	Result
1. Account Registration and Sign-in			
1	Create a new account (register) with a new username and password.	The account is created and the app confirms it.	Pass
2	Register with a username that already exists, or leave the fields blank.	The app rejects it and shows an error message.	Pass
3	Sign in with the new account's correct username and password.	The app signs in and opens the home / dashboard screen.	Pass
4	Sign in with a correct username but a wrong password.	The app rejects the sign-in and shows an error message.	Pass
5	Leave the username and/or password blank and press Sign in.	The app does not sign in and shows an error message.	Pass
2. Prediction Output			
6	Enter 1 kg of calamansi and submit.	Predicted juice yield is shown for the three regression models.	Pass

7	Enter 5 kg and submit.	Predicted yield is shown and is larger than the 1 kg result.	Pass
8	Enter 10 kg and submit.	Predicted yield is shown for all three models.	Pass
9	Enter 500 g (grams) and submit.	The weight is converted and a prediction is shown.	Pass
3. Input Validation and Error Handling			
10	Leave the weight blank and submit.	The app shows an error message and does not predict.	Pass
11	Enter a letter or symbol instead of a number.	The app rejects the input and shows an error message.	Pass
12	Enter 0 or a negative number.	The app rejects the input and shows an error message.	Pass
13	Enter a very large weight (e.g., 1000 kg).	The app shows a result or a clear message, without crashing.	Pass
4. Performance			
14	Check the time between pressing submit and seeing the result.	The result appears in under 2 seconds.	Pass

Note. Number of testers = 1.

Table 13 shows that all 14 test actions passed. The tester was able to register an account, was blocked when using an existing or blank username, and could sign in with correct credentials while wrong or blank credentials were rejected (items 1 to 5). The application returned predictions for 1 kg, 5 kg, and 10 kg, the 5 kg result was larger than the 1 kg result, and a weight of 500 g was converted before predicting (items 6 to 9). Blank, non-numeric, zero, and negative inputs were rejected with an error message, and a very large weight of 1000 kg did not crash the application (items 10 to 13). The result appeared in under 2 seconds, which meets the response time target of the study (item 14).

Table 14. Summary of Beta Testing Results per Category

Category	Test Actions	Passed	Failed	Pass Rate
Account Registration and Sign-in	5	5	0	100%
Prediction Output	4	4	0	100%
Input Validation and Error Handling	4	4	0	100%
Performance	1	1	0	100%
Overall	**14**	**14**	**0**	**100%**

Note. Pass rate = number passed ÷ number of test actions.

As shown in Table 14, the application passed 14 of 14 test actions (100%), and none of the four categories had a failed action.



(1) **(2)**

Figure 24. Blackbox Testing with Tester 1(alpha) 2 (Beta)

4.5.5 White-Box Testing Results (Developer Testing)

White-box testing verified the internal structural logic, conditional branches, mathematical calculations, and exception handling within the application source code (PredictionService, AuthService, and HomeScreen).Verified by the developer.

Table 15: White-Box Structural Test Matrix

Test Case ID	Class / Module Under Test	Target Logic/ Code Branch	Condition / Input Tested	Expected Internal Execution	Observed Result	Status
WB-01	PredictionService	Unit conversion logic	Input in kg vs. g ( )		Converted float equals	**Passed**
WB-02	PredictionService	Batch count estimation	Input		Rounded integer equals	**Passed**
WB-03	PredictionService	SLR formula evaluation	Single fruit ( )		Returns exact expected float	**Passed**
WB-04	PredictionService	Polynomial formula pipeline		Evaluates quadratic & interaction vector	Output matches trained .json coefficients	**Passed**

WB-05	PredictionService	Guard against non-positive entries		Throws ArgumentError exception	Caught gracefully before arithmetic	**Passed**
WB-06	DatabaseService	Supabase RLS query wrapper	Anonymous vs. Authenticated session	Blocks unauthorized table writes	Enforces user UID ownership	**Passed**

All 6 structural test cases achieved 100% path coverage without unhandled exceptions or numerical overflow errors.

4.5.6 User Acceptance Testing

The UAT questionnaire contained 13 statements that cover the functional suitability, usability, performance efficiency, reliability, and usefulness of the application. Each statement was rated from 1 (Strongly Disagree) to 5 (Strongly Agree). The weighted mean of each statement was computed as the sum of the ratings divided by the number of ratings, and was interpreted using the scale in Table 16. The mean of each category is the average of the means of its statements.

Table 16. Interpretation Scale for the Mean Ratings

Mean Range	Verbal Interpretation
4.21 – 5.00	Strongly Agree
3.41 – 4.20	Agree
2.61 – 3.40	Neutral
1.81 – 2.60	Disagree
1.00 – 1.80	Strongly Disagree

Note. Weighted mean = sum of ratings ÷ number of ratings.

Table 17. User Acceptance Testing Results

Statement	5	4	3	2	1	Mean	Interpretation
1. Functional Suitability							
1.1 The app accepts the total weight of calamansi in kilograms or grams.	1	0	0	0	0	5.00	Strongly Agree
1.2 The app displays the predicted juice yield after I enter the weight.	1	0	0	0	0	5.00	Strongly Agree
1.3 The app shows the predictions of the three regression models.	1	0	0	0	0	5.00	Strongly Agree
1.4 The predicted juice yield is reasonable compared with my own experience.	0	0	1	0	0	3.00	Neutral
2. Usability							
2.1 The app is easy to learn and use.	0	1	0	0	0	4.00	Agree
2.2 The screens and buttons are clear and easy to understand.	0	1	0	0	0	4.00	Agree
2.3 The results are presented clearly.	1	0	0	0	0	5.00	Strongly Agree
3. Performance Efficiency and Reliability							
3.1 The app gives results quickly.	1	0	0	0	0	5.00	Strongly Agree
3.2 The app works without errors or crashing.	1	0	0	0	0	5.00	Strongly Agree
4. Usefulness and Acceptance							
4.1 The app can help me plan the purchase of calamansi.	0	1	0	0	0	4.00	Agree
4.2 The app can help me plan juice production.	0	0	1	0	0	3.00	Neutral
4.3 Saved predictions are useful for my records.	0	1	0	0	0	4.00	Agree
4.4 Overall, I accept the app and would use it in my business.	0	1	0	0	0	4.00	Agree

Note. n = 1. With one respondent, the mean of each statement equals the rating given.

As shown in Table 17, the respondent strongly agreed that the application accepts the batch weight in kilograms or grams, displays the predicted juice yield, and shows the

predictions of the three regression models (5.00 each). The respondent rated the statement that the predicted juice yield is reasonable compared with his or her own experience as Neutral (3.00), which was the lowest rating in this category. For usability, the respondent agreed that the application is easy to learn and use and that its screens and buttons are clear (4.00 each), and strongly agreed that the results are presented clearly (5.00). The respondent strongly agreed that the application gives results quickly and works without errors or crashing (5.00 each). For usefulness, the respondent agreed that the application can help in planning the purchase of calamansi (4.00), that saved predictions are useful for records (4.00), and that he or she accepts the application and would use it in the business (4.00), while the statement on planning juice production was rated Neutral (3.00).

Table 18. Summary of User Acceptance Testing Results per Category

Category	Number of Items	Total Ratings	Weighted Mean	Interpretation
Functional Suitability	4	4	4.50	Strongly Agree
Usability	3	3	4.33	Strongly Agree
Performance Efficiency and Reliability	2	2	5.00	Strongly Agree
Usefulness and Acceptance	4	4	3.75	Agree
Overall	**13**	**13**	**4.31**	**Strongly Agree**

Note. Weighted mean of a category = average of the weighted means of its statements. The overall mean is computed from all 13 ratings.

Table 18 shows an overall weighted mean of 4.31 from 13 ratings, interpreted as Strongly Agree. Performance Efficiency and Reliability was rated highest (5.00) and Usefulness and Acceptance was rated lowest (3.75, Agree). The lowest-rated statements

were the reasonableness of the predicted yield compared with the respondent's experience and the usefulness of the application in planning juice production (3.00 each). No statement was rated below 3.

Open-Ended Responses

In the open-ended questions, the respondent liked most that the application is easy to use and gives results quickly. The respondent reported one problem: the predicted calamansi yield was higher than the respondent's actual yield, by at least 14% based on the respondent's regular processing. This agrees with the limitation of the study that the models were trained on juice obtained by manual extraction, while processors typically use extraction machines, so actual yields may differ. The respondent suggested capturing a larger amount of data, covering six months to one year of processing and yield records by type of calamansi and including the supplier and source (area), and adding options to segregate records by source, date processed, and history. These suggestions are considered in the recommendations of the study.

These results assess the observed functionality and acceptance of the application only. They do not independently establish the accuracy of the regression models, which is reported through R^2, MAE, and RMSE, and the predictions are estimates that should not be the sole basis for buying, pricing, or production decisions. Because the results come from a single evaluator, a single beta tester, and a single UAT respondent, they are indicative of the tested functions and cannot be generalized to all juice processors. Overall, the results indicate that the Calamansi Yield Predictor performed its tested functions as intended from the perspective of the evaluator, the tester, and the respondent.

CHAPTER 5

Conclusions and Recommendations

This chapter presents the conclusions synthesized from the development, testing, and evaluation of the Mobile Calamansi Juice Yield Prediction System. It outlines how each research objective was achieved and provides actionable recommendations for juice processors and future researchers.

5.1 Conclusions

Based on the empirical findings, model evaluations, and system testing, the researchers draw the following conclusions:

5.1.1 Fulfillment of Specific Objective 1 (Dataset Collection and Size Categorization)

The study gathered and categorized 1,522 individual calamansi samples from local markets vendors and supplier in Tacloban City, Leyte, with zero missing entries. The physical distribution comprised 287 Small fruits (18.86%), 828 Medium fruits (54.40%), and 407 Large fruits (26.74%), with an empirical mean fruit weight of 12.65 grams. This confirmed that commercial calamansi harvests in the region are predominantly medium-sized.

5.1.2 Fulfillment of Specific Objective 2 (Ground-Truth Juice Extraction and Measurement)

Through digital scale measurement and standardized manual syringe extraction, ground-truth juice volumes were established for all 1,522 samples. The overall mean juice yield was 5.22 mL per fruit, an empirical juice extraction ratio of 41.26%. This laboratory data served as the ground truth for training the models.

5.1.3 Fulfillment of Specific Objective 3 (Model Training, Comparison, and Accuracy Benchmark)

Three regression algorithms were trained and evaluated on an 80:20 data split. Polynomial Regression (Degree 2) emerged as the most accurate model, with R^2 = 0.7102, MAE = 0.5279 mL, RMSE = 0.6637 mL, and MAPE = 10.39% on the test set. The initial target was an R^2 of 0.80, which was not reached. The researchers attribute the lower value to the natural biological variance in citrus crops, such as skin thickness and moisture variations. An MAE of 0.5279 mL, roughly 10 drops of juice per fruit, shows that the model is accurate enough for practical batch forecasting. The comparison also showed that Simple Linear Regression (R^2 = 0.7099) and Multiple Linear Regression (R^2 = 0.7100) perform almost identically, which the researchers attribute to the high correlation between fruit mass and physical size. This indicates that weight is the primary physical driver of juice volume.

5.1.4 Fulfillment of Specific Objective 4 (Cross-Platform Mobile Application Development)

A functional mobile application was developed using Flutter and Dart, backed by Supabase for cloud-synchronized storage. The application lets juice processors enter total batch weight in kilograms or grams, computes comparative predictions across all three algorithms, highlights Polynomial Regression as the optimal model, and keeps an auditable prediction log. Role-based access control separates regular user workflows from the administrator monitoring dashboards.

5.1.5 Fulfillment of Specific Objective 5 (System Performance and Usability Evaluation)

The application delivered a mean end-to-end response time of 337.2 milliseconds, well within the 2.0-second target. Developer white-box testing, external alpha testing by an IT professional, and beta testing with a juice-business partner all recorded 100% pass rates, covering input validation, boundary checking, and interface rendering. One juice-business respondent also rated 13 User Acceptance Testing statements, with an overall weighted mean of 4.31 (Strongly Agree). Usefulness and Acceptance was rated lowest, at 3.75 (Agree). The respondent reported that the predicted yield ran higher than the actual yield from regular machine processing, by at least 14%. This points the same way as the field validation, in which machine extraction yielded about 87.1% of the predicted yield across three commercial batches, an average gap of 12.9%. Together these results give processors a practical adjustment factor for estimating commercial output.

Because the testing involved one evaluator, one beta tester, and one respondent, the results indicate how the tested functions behaved and cannot be generalized to all juice processors.

In summary, the project moved calamansi yield estimation from destructive, subjective manual guessing to an accessible, instant, non-destructive digital forecasting tool for Filipino juice processors.

5.2 Recommendations

Based on the operational insights gained during the design, development, and testing of the system, the researchers propose the following recommendations:

5.2.1 Recommendations for Calamansi Juice Processors

- Adopt Non-Destructive Pre-Extraction Estimation: Processors should integrate the mobile application into their intake stations to estimate juice volumes before purchasing, ensuring fair pricing and preventing financial losses from low-yielding batches.
- Apply the Mechanical Press Loss Factor: In commercial operations using mechanical drum or screw presses, operators should multiply the predicted laboratory yield by the established mechanical efficiency factor (0.871) to estimate bottled inventory.
- Maintain Batch History for Supplier Benchmarking: Processors should use the prediction history log to identify which local growers or seasons consistently deliver higher juice recovery rates per kilogram.

5.2.2 Recommendations for Future Researchers and Developers

- Hardware Scale Integration via Bluetooth (BLE): Future iterations should connect directly to digital kitchen and platform scales through Bluetooth Low Energy, so that batch weights fill the input field automatically without manual typing.
- Computer Vision and Mobile Camera Sizing: While the current system intentionally avoids camera processing to maximize hardware compatibility, future work could explore lightweight on-device edge AI (e.g., TensorFlow Lite) to estimate size distribution from a top-down photo of a batch.
- Multi-Regional and Seasonal Dataset Expansion: The training dataset should be expanded across multiple provinces (e.g., Oriental Mindoro, Quezon, Isabela) and across seasons (dry vs. rainy) to account for seasonal variations in fruit rind thickness and moisture.
- Extraction Machine Adjustment Presets: Future software updates could let processors select their extraction method (Manual Squeeze, Centrifugal Extractor, Screw Press) and automatically apply the matching mechanical efficiency coefficient to the displayed prediction.

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